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United States Patent	12411493
Kind Code	B2
Date of Patent	September 09, 2025
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Mobile machine, control unit, and method of controlling operation of a mobile machine

Abstract

A mobile machine movable between multiple rows of trees includes one or more sensors to output sensor data indicating a distribution of objects in a surrounding environment of the mobile machine, a storage to store environment map data indicating a distribution of trunks of the multiple rows of trees, a localization processor, and a controller to control movement of the mobile machine in accordance with a position of the mobile machine estimated by the localization processor. The localization processor is configured or programmed detect the trunks of the rows of trees in the surrounding environment of the mobile machine based on the sensor data that is repeatedly output from the one or more sensors while the mobile machine is moving, and perform matching between the detected trunks of the rows of trees and the environment map data to estimate a position of the mobile machine.

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Appl. No.: 18/198730

Filed: May 17, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230288936 A1	Sep. 14, 2023

Foreign Application Priority Data

JP	2020-192013	Nov. 18, 2020
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Related U.S. Application Data

continuation parent-doc WO PCT/JP2021/040298 20211101 PENDING child-doc US 18198730

Publication Classification

Int. Cl.: G05D1/00 (20240101); A01B69/04 (20060101); G01C21/30 (20060101)

U.S. Cl.:

CPC G05D1/0214 (20130101); G01C21/30 (20130101); G05D1/024 (20130101); G05D1/0274 (20130101); A01B69/008 (20130101); G05D1/027 (20130101)

Field of Classification Search

CPC: G05D (1/0214); G05D (1/024); G05D (1/0274); G05D (1/027); G05D (1/0088); G05D (1/0231); G05D (1/24); G05D (1/242); G05D (1/246); G05D (1/2462); G05D (1/2465); G05D (2105/15); G05D (2107/21); G05D (2107/22); G05D (2111/14); G05D (2111/17); G01C (21/30); G01C (21/005); G01C (21/38); G01C (21/3804); A01B (69/008); A01B (69/001); G01S (17/89); G01S (17/931); B60R (2021/0076); A01G (23/00); G06T (7/73); G06T (7/75); G06V (20/56)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority to Japanese Patent Application No. 2020-192013 filed on Nov. 18, 2020 and is a

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present disclosure relates to a mobile machine, a control unit, and a method of controlling operation of a mobile machine.

2. Description of the Related Art

(2) As attempts in next-generation agriculture, research and development of smart agriculture utilizing ICT (Information and Communication Technology) and IoT (Internet of Things) is under way. Research and development are also directed to the automation and unmanned use of tractors or other work vehicles to be used in the field. For example, work vehicles which travel via automatic steering by utilizing a positioning system that is capable of precise positioning, e.g., a GNSS (Global Navigation Satellite System), are coming into practical use. International Publication No. 2017/208306, Japanese Laid-Open Patent Publication No. 2020-104617 and Japanese Laid-Open Patent Publication No. 2020-12680 disclose examples of work vehicles that perform automatic steering based on positioning results obtained by using a GNSS.

(3) On the other hand, development of mobile machines which autonomously move by utilizing distance sensors, e.g., LiDAR (Light Detection and Ranging) is also under way. For example, Japanese Laid-Open Patent Publication No. 2019-154379 discloses an example of a work vehicle which performs self-driving in between crop rows in a field by utilizing LiDAR.

SUMMARY OF THE INVENTION

(4) In an environment in which trees are distributed with a high density, e.g., vineyards or other orchards or forests, leaves thriving in upper portions of the trees create canopies, each of which serves as an obstacle or a multiple reflector against radio waves from a satellite. Such an environment hinders accurate positioning using a GNSS. In an environment where a GNSS cannot be used, use of SLAM (Simultaneous Localization and Mapping), where localization and map generation simultaneously take place, might be possible. However, various challenges exist in the practical application of a mobile machine that uses SLAM to move autonomously or with automatic steering in an environment with a multitude of trees. One challenge is that the distribution of tree leaves changes significantly with seasonal changes, making it impossible to continue using maps that were created in the past, for example.

(5) A mobile machine according to an illustrative preferred embodiment of the present disclosure is movable between multiple rows of trees. The mobile machine includes one or more sensors to output sensor data indicating a distribution of objects in a surrounding environment of the mobile machine, a storage to store environment map data indicating a distribution of trunks of the multiple rows of trees, a localization processor, and a controller to control movement of the mobile machine in accordance with a position of the mobile machine estimated by the localization processor. The localization processor is configured or programmed to detect the trunks of the rows of trees in the surrounding environment of the mobile machine based on the sensor data that is repeatedly output from the one or more sensors while the mobile machine is moving, and perform matching between the detected trunks of the rows of trees and the environment map data to estimate a position of the mobile machine.

(6) General or specific aspects of preferred embodiments of the present disclosure and modifications or combinations thereof may be implemented using a device, a system, a method, an integrated circuit, a computer program, a computer-readable recording medium, or any combination thereof. The computer-readable recording medium may be inclusive of a volatile recording medium, or a non-volatile recording medium. The device may include a plurality of devices. In the case where the device includes two or more devices, the two or more devices may be disposed within a single apparatus, or divided over two or more separate apparatuses.

(7) According to a preferred embodiment of the present disclosure, even in an environment where positioning by GNSS is difficult, automatic steering or autonomous movement of a mobile machine can be achieved. A distribution of the trunks of the rows of trees undergoes less seasonal change than does a distribution of leaves. By generating an environment map while focusing on the trunks, it becomes possible to continuously use the same environment map over relatively periods of time.

(8) The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the preferred embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram schematically showing an example of a mobile machine that moves on a plane from time t_1 to time t_2 .
- (2) FIG. 2A is a diagram showing a mobile machine and a plurality of landmarks at time t_1 .
- (3) FIG. 2B is a diagram showing a mobile machine and a plurality of landmarks at time t_2 .
- (4) FIG. 3 is a diagram showing a placement relationship between a mobile machine coordinate system $\Sigma_{sub.V}$ and a world coordinate system $\Sigma_{sub.W}$.
- (5) FIG. 4 is a diagram showing an example of matching.
- (6) FIG. 5 is a diagram showing an example of matching.
- (7) FIG. 6 is a perspective view showing the appearance of a tractor according to Preferred Embodiment 1 of the present invention.
- (8) FIG. 7 is a schematic diagram showing a tractor as viewed in a lateral direction.
- (9) FIG. 8 is a block diagram showing an exemplary schematic configuration of a tractor and an implement.
- (10) FIG. 9A is a schematic diagram of a LiDAR sensor as viewed in a lateral direction of the tractor.
- (11) FIG. 9B is a schematic diagram of a LiDAR sensor as viewed from vertically above.
- (12) FIG. 10 is a block diagram showing an exemplary schematic configuration of a LiDAR sensor.
- (13) FIG. 11 is a diagram schematically showing an example of an environment in which a tractor travels.
- (14) FIG. 12 is a perspective view schematically showing the surrounding environment of a tractor.
- (15) FIG. 13A is a diagram schematically showing an example of a travel path of a tractor.
- (16) FIG. 13B is a diagram schematically showing another example of a travel path of a tractor.
- (17) FIG. 14A is a diagram schematically showing a tree and the ground surface being irradiated with laser beams emitted from the LiDAR sensor of the tractor.
- (18) FIG. 14B is a diagram schematically showing the ground surface being irradiated with laser beams emitted from the LiDAR sensor.
- (19) FIG. 15A is a diagram schematically showing an example of a tractor and a distribution of reflection points of laser pulses radiated from the LiDAR sensor at a certain point in time.
- (20) FIG. 15B is a diagram schematically showing an example of a tractor and a distribution of reflection points of laser pulses radiated from the LiDAR sensor at a point in time slightly later than the point in time of FIG. 15A.
- (21) FIG. 16 is a diagram schematically showing an example point cloud of trunks obtained through one scan of laser beams in multiple layers.
- (22) FIG. 17A is a diagram schematically showing a point cloud of trunks being obtained by a scan of laser beams in a layer $L_{sub.k}$, $L_{sub.k+1}$, $L_{sub.k+2}$.
- (23) FIG. 17B is a diagram schematically showing a point cloud of trunks being obtained by a scan of laser beams in a layer $L_{sub.k+1}$.

(24) FIG. 17C is a diagram schematically showing a point cloud of trunks being obtained by a scan of laser beams in a layer $L_{sub.k+2}$.

(25) FIG. 18 is a graph showing an example relationship between distances to reflection points that are measured by laser beams in multiple layers through one scan, and azimuth angles of the reflection points.

(26) FIG. 19 is a diagram showing an example of a local map generated by extracting reflection points that are located on the surface of trunks from scan data that is acquired by a tractor being at a certain position.

(27) FIG. 20A is a plan view schematically showing a process of estimating the position of a mobile machine by matching a local map to an already-established environment map.

(28) FIG. 20B is a plan view schematically showing a process of estimating the position of a mobile machine by matching a local map to an already-established environment map.

(29) FIG. 21A is a diagram showing an example where a LiDAR sensor is attached to the lower front of a vehicle body of the tractor.

(30) FIG. 21B is a diagram showing an example where a LiDAR sensor is attached to the upper front of a cabin of the tractor.

(31) FIG. 22 is a diagram showing an exemplary functional configuration of a localization module.

(32) FIG. 23 is a diagram schematically showing an example format of environment map data.

(33) FIG. 24 is a diagram showing an example temporal relationship between IMU data and scan data.

(34) FIG. 25 is a flowchart showing a localization operation.

(35) FIG. 26 is a diagram for describing a filtering process for scan data.

(36) FIG. 27 is a flowchart showing an example control to be performed after the pose of a tractor has been estimated.

(37) FIG. 28A is a diagram showing an example of a tractor traveling along an intended travel path P.

(38) FIG. 28B is a diagram showing an example of a tractor at a position which is shifted rightward from the intended travel path P.

(39) FIG. 28C is a diagram showing an example of a tractor at a position which is shifted leftward from the intended travel path P.

(40) FIG. 28D is a diagram showing an example of a tractor which is oriented in an inclined direction with respect to the intended travel path P.

(41) FIG. 29A is a diagram showing an example of two-dimensional data indicating a trunk distribution of rows of trees that is generated based on the environment map.

(42) FIG. 29B is a diagram for describing an example method of determining an intended travel path based on trunk distribution data.

(43) FIG. 30 is a diagram showing another example of trunk distribution data for path setting purposes.

(44) FIG. 31A is a diagram schematically showing an example environment in which rows of trees are in a curved placement.

(45) FIG. 31B is a diagram schematically a point cloud that may be observed through a single scan in the environment shown in FIG. 31A.

(46) FIG. 31C is a diagram for describing an operation of determining a travel path by estimating the positions of hidden trunks.

(47) FIG. 31D is a diagram showing an example of grid data indicating a two-dimensional distribution of trunks in the case where rows of trees are in a curved placement.

(48) FIG. 32 is a diagram schematically showing an example situation where an obstacle and a person exist in a traveling environment of a tractor.

(49) FIG. 33A is a diagram showing an example placement of the trunks of rows of trees detected by a tractor.

- (50) FIG. 33B is a diagram showing an example trunk distribution indicated by environment map data.
- (51) FIG. 34 is a schematic diagram of a tractor according to Preferred Embodiment 2 as viewed in a lateral direction.
- (52) FIG. 35 is a block diagram showing an exemplary schematic configuration of a tractor according to Preferred Embodiment 2 of the present invention.
- (53) FIG. 36A is a diagram schematically showing a tractor traveling near a tree.
- (54) FIG. 36B is another diagram schematically showing a tractor traveling near a tree.
- (55) FIG. 36C is still another diagram schematically showing a tractor traveling near a tree.
- (56) FIG. 37 is a diagram schematically showing an exemplary configuration of a system according to Preferred Embodiment 3 of the present invention.
- (57) FIG. 38 is a block diagram showing an exemplary configuration of a system according to Preferred Embodiment 3 of the present invention.
- (58) FIG. 39 is a perspective view showing an example appearance of a drone.
- (59) FIG. 40 is a perspective view schematically an example manner of data collecting work by multiple drones.
- (60) FIG. 41 is a diagram schematically showing a tree being irradiated with laser beams emitted from a LiDAR sensor mounted on a drone.
- (61) FIG. 42 is a diagram showing an example format of local map data to be output from a drone.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (62) Definitions of main terms used in the present specification are described below.
- (63) A “mobile machine” is a device equipped with wheels, crawlers, bipedal or multipedal walking devices, or propellers or other driving devices that generate traction for movement. “Movable units”, as used in the present disclosure, include work vehicles such as tractors, transport vehicles, mobile robots, and unmanned aerial vehicles (UAVs, so-called drones) such as multicopters. Movable units may be unmanned or manned.
- (64) A “work vehicle” means a vehicle capable of traveling while performing a specific task in a work area such as a field (e.g., orchard, field, paddy field, or pasture), mountain forest, or construction site. Examples of work vehicles include agricultural machines such as tractors, rice transplanters, combines, vehicles for crop management, and riding mowers, as well as vehicles used for non-agricultural purposes, such as construction vehicles and snowplow vehicles.
- (65) “SLAM” is a generic term for techniques where localization of a mobile machine and map generation simultaneously take place.
- (66) “Localization” is the estimation of the position of a mobile machine on a map (e.g., the position of the center of gravity of the mobile machine). In a localization based on SLAM, usually, a pose of the mobile machine is determined.
- (67) A “pose” is the “position and orientation” of an object. A pose in a two-dimensional space is defined by three coordinate values (x,y, θ), for example. Herein, (x,y) are coordinate values of an XY coordinate system, which is a world coordinate system that is fixed to the globe, and θ is an angle relative to a reference direction. A pose in a three-dimensional space is defined by six coordinate values (x,y,z, $\theta_{\text{sub.R}}$, $\theta_{\text{sub.P}}$, $\theta_{\text{sub.Y}}$), for example. Herein, (x,y,z) are coordinate values of an XYZ coordinate system which is a world coordinate system, and ($\theta_{\text{sub.R}}$, $\theta_{\text{sub.P}}$, $\theta_{\text{sub.Y}}$) are angles of roll, pitch, and yaw relative to respective reference directions. The attitude of a mobile machine is expressed as ($\theta_{\text{sub.R}}$, $\theta_{\text{sub.P}}$, $\theta_{\text{sub.Y}}$). A roll angle $\theta_{\text{sub.R}}$ represents the amount of rotation of the mobile machine around its front-rear axis. A pitch angle $\theta_{\text{sub.P}}$ represents the amount of rotation of the mobile machine around its right-left axis. A yaw angle $\theta_{\text{sub.Y}}$ represents the amount of rotation of the mobile machine around its top-bottom axis. The attitude may be defined by an Euler angle or other angles, or a quaternion.
- (68) “Environment map data” is data based on a predetermined coordinate system that expresses positions or regions of objects within an environment in which a mobile machine moves.

Environment map data may simply be referred to as an “environment map”. Examples of coordinate systems defining an environment map include not only world coordinate systems such as a geographic coordinate system that is fixed to the globe, but also odometry coordinate systems indicating poses based on odometry information, and so on. Environment map data may include information other than position (e.g., attribute or other information) of an object existing in an environment. The environment map encompasses maps of various forms, such as point cloud maps or grid maps. Hereinafter, an environment map may be referred to as “map data” or simply as a “map”. Moreover, the data of a local map or partial map that is generated or processed in the course of establishing an environment map may also be referred to as “map data”, or simply as a “map”.

(69) “Automatic steering” means the steering of a mobile machine that is based on the action of a controller, rather than manually. In the present specification, “automatic travel” is a notion that encompasses “automatic steering”. A portion or an entirety of the controller may be external to the mobile machine. Communications of control signals, commands, data, or the like may be performed between the mobile machine and the controller external to the mobile machine. During automatic steering, other operations such as velocity control may be performed manually.

(70) “Autonomous movement” means the movement of a mobile machine being made based on the action of a controller while sensing the surrounding environment, without any person being involved in the control of the movement. Autonomous movement includes autonomous driving and autonomous flight. The controller may control the necessary movements of the mobile machine, such as steering, velocity control, and starting and stopping of travel, as well as ascending, descending, and hovering during flight. Autonomous movement can include obstacle detection and obstacle avoidance actions. Autonomous movement may include operations of obstacle detection and obstacle avoidance.

(71) “Self-driving” encompasses autonomous driving based on a controller which is included within the mobile machine, and also traveling based on commands coming from a computer in an operating schedule management system. Autonomous driving includes not only a movement of the mobile machine toward a destination along a predetermined path, but also a movement of merely following a target of tracking. Moreover, it may also be possible to temporarily move based on instructions from a human worker.

(72) A “localization device” or “localization processor” is a device or a processor that estimates its own position on an environment map, based on sensor data that is acquired by an external sensor, such as a LiDAR (Light Detection and Ranging) sensor.

(73) An “external sensor” is a sensor that senses the external state of the mobile machine. Examples of external sensors include laser range finders (also referred to as “range sensors”), cameras (or image sensors), LiDAR sensors, millimeter wave radars, and magnetic sensors.

(74) An “internal sensor” is a sensor that senses the state of the mobile machine. The internal sensor includes a wheel encoder to measure the rotational speed of a wheel, an acceleration sensor, and an angular acceleration sensor (e.g., a gyroscope). An inertial measurement unit (IMU) includes an acceleration sensor and an angular acceleration sensor, and is able to output an amount of move and an attitude of the mobile machine. Information representing amount of change in the pose of the mobile machine that is acquired by the internal sensor is referred to as “odometry information”.

(75) The “trunk of a tree” is a lignified stem of a woody plant that is the main axis which stands upright above the ground surface and produces branches. It does not include the branches, leaves, and the root of the tree.

(76) Next, fundamental principles of localization that utilize SLAM technique used in a preferred embodiment of the present disclosure will be described. For simplicity, it is assumed herein that the mobile machine moves in a two-dimensional space (i.e., a plane).

(77) First, FIG. 1 is referred to. FIG. 1 is a diagram schematically showing an example of a mobile machine 100 that moves on a plane from time t_1 to time t_2 . In this example, the mobile machine

100 is a work vehicle (e.g., a tractor) that travels on wheels. The mobile machine **100** may be a vehicle other than a tractor, or any other kind of mobile machine, such as a walking robot or a drone. FIG. 1 schematically shows the position and orientation of the mobile machine **100** moving on a plane. The position coordinates of the mobile machine **100** are indicated by an XY coordinate system. The XY coordinate system is a world coordinate system $\Sigma_{\text{sub.W}}$ that is fixed to the globe. FIG. 1 also shows a mobile machine coordinate system Σ_V that is fixed to the mobile machine **100**. In this example, the mobile machine coordinate system Σ_V is a uv coordinate system whose u-axis direction is the direction in which the front of the mobile machine **100** faces and whose v-axis direction is a direction resulting from rotating the u axis counterclockwise by 90 degrees. The origin position of the mobile machine coordinate system $\Sigma_{\text{sub.V}}$ changes with movement of the mobile machine **100**. The orientation of the mobile machine coordinate system $\Sigma_{\text{sub.V}}$ changes with change in the orientation of the mobile machine **100**, i.e., rotation.

(78) Assume that the mobile machine **100** has a pose r_1 (i.e., position and orientation) at time t_1 , and a pose r_2 at time t_2 . The pose r_1 is defined by a position indicated by coordinates (x_1, y_1) and an orientation indicated by an angle θ_1 , for example. It is assumed herein that the orientation of the mobile machine **100** is the direction of the front of the mobile machine **100**. It is further assumed that the positive direction on the X axis defines a reference direction of angle, and that the counterclockwise direction defines the positive direction of angle. The pose r_2 is defined by a position indicated by coordinates (x_2, y_2) and an orientation indicated by an angle θ_2 .

(79) In the example of FIG. 1, from time t_1 to time t_2 , the position of the mobile machine **100** moves (translates) in the direction of the angle θ_1 by Δd_1 , and the orientation of the mobile machine **100** rotates counterclockwise by $\Delta\phi_1 (= \theta_2 - \theta_1)$. Thus, the motion of the mobile machine **100** is a synthesis of “translation” and “rotation”.

(80) In this example, when the distance Δd_1 is sufficiently short, the traveling direction of the mobile machine **100** can be approximated as being parallel to the u axis of the mobile machine coordinate system $\Sigma_{\text{sub.V}}$. Therefore, the following eq. 1 holds true.

$$(81) \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} \cos \theta_1 & -\sin \theta_1 \\ \sin \theta_1 & \cos \theta_1 \end{pmatrix} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} \Delta d_1 \cos \theta_1 \\ \Delta d_1 \sin \theta_1 \end{pmatrix} \quad (\text{eq. 1})$$

(82) When the mobile machine **100** includes an internal sensor(s) such as a revolutions sensor for a wheel and/or an inertial measurement unit (IMU), it is possible to acquire estimated values of Δd_1 and $\Delta\phi_1$ from such internal sensors, i.e., odometry information. If the time span from time t_1 to time t_2 is as short as e.g., about 10 milliseconds, the distance Δd_1 is sufficiently short, and eq. 1 holds true. As time proceeds from t_1 to $t_2, t_3, \dots$, the estimated values of Δd_1 and $\Delta\phi_1$ are periodically updated, such that changes in the position and orientation (i.e., pose) of the mobile machine **100** can be estimated. In other words, if the initial pose, e.g. (x_1, y_1, θ_1) , is known, the estimated value of any subsequent pose of the mobile machine **100** can be periodically updated based on odometry information. However, pose estimation based on odometry information has the problem of accumulated errors. Therefore, in many cases, it is necessary to acquire highly-accurate estimated values of the position of the mobile machine **100** by utilizing a satellite positioning system or SLAM technique.

(83) Next, with reference to FIG. 2A to FIG. 5, a localization that is performed by using an external sensor such as a LiDAR will be described. In the following, examples of localization operations will be described with respect to the case where environment map data exists and the case where environment map data does not exist.

(84) In the examples shown in FIGS. 2A and 2B, in the environment in which the mobile machine **100** moves from time t_1 to time t_2 , landmarks m_1, m_2, m_3 and m_4 of fixed positions exist. In this example, environment map data indicating the positions of the landmarks already exists. Such environment map data includes an identifier (e.g., a landmark number) that identifies each

individual landmark, and position coordinates of the landmark associated with each identifier in the world coordinate system $\Sigma_{sub.W}$, for example. The coordinates of the landmarks m_1 , m_2 , m_3 and m_4 in the world coordinate system $\Sigma_{sub.W}$ are assumed to be, respectively, $(x_{sub.m1}, y_{sub.m1})$, $(x_{sub.m2}, y_{sub.m2})$, $(x_{sub.m3}, y_{sub.m3})$ and $(x_{sub.m4}, y_{sub.m4})$. Each point in point cloud data that is acquired by LiDAR may serve as a landmark. An environment map that includes point cloud data is referred to as a point cloud map. The landmark positions may be indicated by cells on a grid map.

(85) In the example of FIG. 2A, observed values (or measured values) of the landmarks m_1 , m_2 , m_3 and m_4 acquired by the vehicle **100** at the pose r_1 are, respectively, z_1 , z_2 , z_3 and z_4 . The observed values are, for example, values representing the distance and direction toward each landmark as measured by external sensors. Since observed values are data that is acquired by external sensors, they may be referred to as “sensor data” in the present disclosure.

(86) As the mobile machine **100** moves as shown in FIG. 2B, the landmarks m_1 , m_2 and m_3 as acquired by the mobile machine **100** at the pose r_2 may take observed values z_5 , z_6 and z_7 , respectively.

(87) Based on these observed values, position coordinates of the landmarks on the mobile machine coordinate system $\Sigma_{sub.V}$ can be obtained. As described above, because the mobile machine coordinate system $\Sigma_{sub.V}$ is a coordinate system that is fixed to the mobile machine **100**, the position coordinates of the same landmark (e.g., the landmark m_1) on the mobile machine coordinate system $\Sigma_{sub.V}$ will change with the changing pose of the mobile machine **100**.

(88) The position of a landmark acquired based on observed values has coordinates in a sensor coordinate system that is determined by the positions and orientations of external sensors. Strictly speaking, a sensor coordinate system may differ from the mobile machine coordinate system $\Sigma_{sub.V}$. In the following description, however, the sensor coordinate system and the mobile machine coordinate system $\Sigma_{sub.V}$ are assumed to be identical. Because the relationship between the sensor coordinate system and the mobile machine coordinate system $\Sigma_{sub.V}$ is known, one coordinate system can be matched to the other coordinate system by rotating the one coordinate system by a known angle and translating the one coordinate system by a known distance.

(89) By observing the plurality of landmarks, the mobile machine **100** moving in an environment acquires the position coordinates of each landmark in the mobile machine coordinate system $\Sigma_{sub.V}$. Then, if the position coordinates of each landmark in the world coordinate system $\Sigma_{sub.W}$ are included in the environment map data, it is possible to estimate the poses r_1 and r_2 based on the observed values z_1 to z_7 , etc. Such estimation is enabled through a matching between the position coordinates of each landmark as determined from observed value and the position coordinates of the landmark included in the environment map data, for example.

(90) Next, with reference to FIG. 3, a method of converting coordinates (u_1, v_1) on the mobile machine coordinate system $\Sigma_{sub.V}$ into coordinates (u_1', v_1') on the world coordinate system $\Sigma_{sub.W}$ will be described. FIG. 3 shows a placement relationship between the mobile machine coordinate system $\Sigma_{sub.V}$ and the world coordinate system $\Sigma_{sub.W}$ at time t_2 . In the example of FIG. 3, for simplicity, the origin of the mobile machine coordinate system $\Sigma_{sub.V}$ is allowed to match the origin (sensor center) of the sensor coordinate system, such that both coordinate systems are identical. The coordinates (u_1, v_1) on the mobile machine coordinate system $\Sigma_{sub.V}$ are calculated based on the observed value z_5 of the landmark m_1 from the mobile machine **100** of the pose r_2 at time t_2 . Assuming that the observed value z_5 is defined by a distance $r_{sub.m1}$ and an angle $\phi_{sub.m1}$ relative to the u axis, then u_1 and v_1 are calculated as $u_1 = r_{sub.m1} \cos \phi_{sub.m1}$, $v_1 = r_{sub.m1} \sin \phi_{sub.m1}$. The conversion (coordinate transformation) from coordinates on the mobile machine coordinate system $\Sigma_{sub.V}$ to coordinates on the world coordinate system $\Sigma_{sub.W}$ is made by rotating the mobile machine coordinate system $\Sigma_{sub.V}$ clockwise by the angle θ_2 and translating the origin of the mobile machine coordinate system $\Sigma_{sub.V}$ to the origin of the world coordinate system $\Sigma_{sub.W}$. At this time, the following eq. 2 and eq. 3 hold true.

$$(91) \begin{pmatrix} u1' \\ v1' \end{pmatrix} = \begin{pmatrix} \cos \theta_2 & -\sin \theta_2 \\ \sin \theta_2 & \cos \theta_2 \end{pmatrix} \begin{pmatrix} u1 \\ v1 \end{pmatrix} + \begin{pmatrix} x2 \\ y2 \end{pmatrix} \quad (\text{eq. 2}) \quad \begin{pmatrix} u1' \\ v1' \end{pmatrix} = R \begin{pmatrix} u1 \\ v1 \end{pmatrix} + T \quad (\text{eq. 3})$$

(92) Herein, R is a rotation matrix that is determined by the orientation of the mobile machine **100**, and T is a position vector of the mobile machine **100** in the world coordinate system $\Sigma_{\text{sub.W}}$. The contents of the rotation matrix R and the position vector T are determined from the pose of the mobile machine **100**.

(93) The $(u1', v1')$ obtained from the coordinate transformation should match $(x_{\text{sub.m1}}, y_{\text{sub.m1}})$, which are the coordinates of the landmark m1 in the world coordinate system $\Sigma_{\text{sub.W}}$. However, if the estimated values of the pose of the mobile machine **100** are deviated from the true values, an error (distance) may occur between $(u1', v1')$ and $(x_{\text{sub.m1}}, y_{\text{sub.m1}})$. To perform localization is to determine the contents of the rotation matrix R and the position vector T in such a manner that the error between the $(u1', v1')$ and $(x_{\text{sub.m1}}, y_{\text{sub.m1}})$ becomes small.

(94) Next, with reference to FIG. 4 and FIG. 5, an example of matching will be described in more detail. FIG. 4 schematically shows the coordinates of the landmarks m1, m2 and m3 on a uv coordinate system which is the mobile machine coordinate system $\Sigma_{\text{sub.V}}$ at time t2. These coordinates may be calculated from the observed values z5, z6 and z7. In this example, the coordinates that are calculated from the observed values z5, z6 and z7 are assumed to be $(u5, v5)$, $(u6, v6)$ and $(u7, v7)$, respectively. By performing a coordinate transformation from the mobile machine coordinate system $\Sigma_{\text{sub.V}}$ to the world coordinate system $\Sigma_{\text{sub.W}}$, $(u5, v5)$, $(u6, v6)$ and $(u7, v7)$ on the mobile machine coordinate system $\Sigma_{\text{sub.V}}$ are converted to, respectively, $(u5', v5')$, $(u6', v6')$ and $(u7', v7')$ on the world coordinate system $\Sigma_{\text{sub.W}}$, as shown in FIG. 5. FIG. 5 shows an example state where a discrepancy exists between the converted coordinates $(u5', v5')$, $(u6', v6')$ and $(u7', v7')$ and the coordinates $(x_{\text{sub.m2}}, y_{\text{sub.m2}})$, $(x_{\text{sub.m3}}, y_{\text{sub.m3}})$ and $(x_{\text{sub.m4}}, y_{\text{sub.m4}})$ of their corresponding landmarks. The coordinate transformation is defined by a rotation based on the rotation matrix R and a translation based on the position vector T. On the world coordinate system $\Sigma_{\text{sub.W}}$, the contents of the coordinate transformation (R, T) are to be determined so that $(u5', v5')$, $(u6', v6')$ and $(u7', v7')$ respectively approximate (i.e., match) $(x_{\text{sub.m2}}, y_{\text{sub.m2}})$, $(x_{\text{sub.m3}}, y_{\text{sub.m3}})$ and $(x_{\text{sub.m4}}, y_{\text{sub.m4}})$, which are the coordinates of their corresponding landmarks.

(95) In eq. 2, the unknowns are x2, y2 and θ_2 . Since there are three unknowns, three or more equations corresponding to eq. 2 shall solve x2, y2 and θ_2 through calculations. When three or more landmarks are observed from the mobile machine **100** of the same pose r2 as shown in FIG. 5, three equations corresponding to eq. 2 are obtained. Thus, by solving these simultaneous equations, calculated values of the pose r2 (x2, y2, θ_2) can be acquired. Since actual observations will involve errors, simultaneous equations based on a multitude of equations, i.e., more than three, are to be used, and optimization by the least-squares method or the like is to be made.

(96) Examples of such algorithms for estimating a vehicle's own position through matching-based optimizations include ICP matching technique and NDT matching technique. Either of these matching techniques, or any other method may be used.

(97) When estimating the position of the mobile machine **100** from observed values and environment map data, it is not always necessary to estimate the orientation of the mobile machine **100**. For example, as shown in FIGS. 2A and 2B, when the distance to a landmark of a known position is able to be acquired from the observed value, the position of the mobile machine **100** (or more precisely, the position of the range sensor) can be calculated based on the principle of triangulation. Estimation of a position based on such principles is also encompassed within "localization".

(98) When environment map data does not exist, it is necessary to perform a process of estimating the pose of the mobile machine **100** while determining the coordinates of the landmarks m1, m2, m3 and m4, . . . based on observed values acquired by the mobile machine **100** moving as shown in

FIGS. 2A and 2B (i.e., estimation). The coordinates of the landmarks m_1 , m_2 , m_3 and m_4 , . . . on the world coordinate system $\Sigma_{sub.W}$ constitute environment map data. Therefore, in order to establish a map, it is necessary to determine the position coordinates, on the world coordinate system $\Sigma_{sub.W}$, of objects to serve as the landmarks.

(99) In the examples of FIGS. 2A and 2B, the moving mobile machine **100** acquires a plurality of observed values from each of the landmarks m_2 , m_3 and m_4 . By making more observations, it is possible to obtain a greater number of simultaneous equations than the total number of unknowns included among the pose of the mobile machine **100** and the landmark coordinates. This allows estimated values of the pose of the mobile machine **100** and landmark coordinates to be calculated.

(100) There are many kinds of algorithms for performing localization and environment map generation based on SLAM. Examples of SLAM algorithms include not only algorithms utilizing LiDAR sensors, but also algorithms utilizing other external sensors such as cameras. Bayesian filters such as particle filters may be used for localization, or graph-based methods may be used to improve the accuracy of pose estimation. In preferred embodiments of the present disclosure, there is no limitation as to the kind of SLAM algorithm.

(101) The LiDAR sensors may include scan-type sensors, which acquire information on the distance distribution of objects in space by scanning a laser beam, and flash-type sensors, which acquire information on the distance distribution of objects in space by using light diffused over a wide area. A scan-type LiDAR sensor uses a higher intensity light than does a flash-type LiDAR sensor, and thus can acquire distance information at a greater distance. On the other hand, flash-type LiDAR sensors are suitable for applications that do not require intense light because they are simple in structure and can be manufactured at low cost. The present disclosure mainly describes examples where a scan-type LiDAR sensor(s) is used, but a flash-type LiDAR sensor(s) may also be used in some applications.

(102) When an object in the environment is to be observed with a typical scan-type LiDAR sensor, a pulsed laser beam (i.e., laser pulses) is emitted, and the time until the laser pulses reflected by an object existing in the surrounding environment return to the LiDAR sensor is measured, such that the distance and direction toward a reflection point that is located on the object surface can be known. Once the distance and direction toward the reflection point are known, the coordinates of the “reflection point” in the mobile machine coordinate system $\Sigma_{sub.V}$ can be acquired. Scan-type LiDAR sensors can be classified into two-dimensional LiDAR sensors and three-dimensional LiDAR sensors. With a two-dimensional LiDAR sensor, the environment may be scanned so that the laser beam rotates within a single plane. On the other hand, with a three-dimensional LiDAR sensor, the environment may be scanned so that each of a plurality of laser beams rotates along a respectively different conical surface. The coordinates (two-dimensional or three-dimensional coordinate values) of each individual reflection point that are acquired with such LiDAR sensors are expressed by the mobile machine coordinate system $\Sigma_{sub.V}$. By converting the coordinates of each individual reflection point from the mobile machine coordinate system $\Sigma_{sub.V}$ to the world coordinate system $\Sigma_{sub.W}$, it becomes possible to obtain coordinates of each individual reflection point on the world coordinate system $\Sigma_{sub.W}$, such that a point cloud map can be established. In order to convert from the mobile machine coordinate system $\Sigma_{sub.V}$ to the world coordinate system $\Sigma_{sub.W}$, as described earlier, information of the pose of the mobile machine **100** is necessary.

(103) Under the SLAM technique, estimation of the pose of the mobile machine **100** and establishment of an environment map can be achieved simultaneously. However, in establishing an environment map, techniques other than the SLAM technique may be used to estimate or measure the pose of the mobile machine **100**. The reason is that the pose of the mobile machine **100** can also be determined by using a satellite positioning system that measures the position of the mobile machine **100** and an inertial guidance device. However, in situations where a positioning system such as a satellite positioning system is not available, it is necessary to estimate the pose of the

mobile machine **100** by using the SLAM technique to establish an environment map. Note that the position of the mobile machine **100** may be estimated by using the SLAM technique, whereas the orientation or attitude of the mobile machine **100** may be estimated by using another sensor such as an inertial measurement unit.

(104) Hereinafter, preferred embodiments of the present disclosure will be described more specifically. Note however that unnecessarily detailed descriptions may be omitted. For example, detailed descriptions of what is well known in the art or redundant descriptions on what is substantially the same configuration may be omitted. This is to avoid lengthy description, and facilitate the understanding of those skilled in the art. The accompanying drawings and the following description, which are provided by the present inventors so that those skilled in the art can sufficiently understand the present disclosure, are not intended to limit the scope of claims. In the following description, elements or features having identical or similar functions are denoted by identical reference numerals.

Preferred Embodiment 1

(105) A mobile machine according to a first preferred embodiment of the present disclosure will be described.

(106) A mobile machine according to the present preferred embodiment is usable in an environment where multiple trees grow to form multiple rows of trees. The mobile machine includes one or more sensors to output sensor data indicating a distribution of objects in a surrounding environment, a storage, a localization processor, and a controller to control movement of the mobile machine. The storage stores environment map data indicating a distribution of trunks of the multiple rows of trees. Based on sensor data that is repeatedly output from one or more sensors while the mobile machine is moving, the localization processor is configured or programmed to detect trunks of the rows of trees in the surrounding environment of the mobile machine, and estimate the position of the mobile machine through matching between the detected trunks of the rows of trees and the environment map data. The controller is configured or programmed to control movement of the mobile machine in accordance with the estimated position of the mobile machine. The mobile machine according to the present preferred embodiment further includes a data generator configured or programmed to generate environment map data, or local map data from which to generate environment map data. While performing localization based on sensor data that is repeatedly output from one or more sensors while the mobile machine is moving, the data generator detects trunks of the rows of trees in the surrounding environment of the mobile machine, and generates local map data, from which to generate environment map data indicating a distribution of the detected trunks of the rows of trees, and stores the local map data to the storage. Also, by joining together repeatedly generated local map data, the data generator may generate environment map data, and record it to the storage. Thus, the mobile machine according to the present preferred embodiment has the function of generating an environment map while moving between multiple rows of trees, and the function of autonomously moving between rows of trees while estimating the position of the mobile machine by utilizing the generated environment map.

(107) The one or more sensors may include at least one LiDAR sensor that outputs two-dimensional or three-dimensional point cloud data as sensor data, for example. In the present specification, “point cloud data” broadly means data indicating a distribution of multiple reflection points that are observed with a LiDAR sensor(s). The point cloud data may include coordinate values of each reflection point in a two-dimensional space or a three-dimensional space and/or information indicating the distance and direction of each reflection point, for example.

(108) The LiDAR sensor(s) repeatedly outputs point cloud data with a pre-designated cycle, for example. The data generator can detect trunks based on the position of each point in the point cloud data that is output during a period of one cycle or longer, or on the distance or angle of each point from the mobile machine.

(109) Hereinafter, the configuration and operation according to the present preferred embodiment

will be described, mainly with respect to an example where the mobile machine is a work vehicle such as a tractor for use in a task to be performed at an orchard such as a vineyard, and where the sensor(s) is a scan-type LiDAR sensor(s).

(110) FIG. 6 is a perspective view showing the appearance of a tractor **100A** according to the present preferred embodiment. FIG. 7 is a schematic diagram of the tractor **100A** as viewed in a lateral direction. FIG. 7 also shows an implement **300** to be linked to the tractor **100A** in use.

(111) As shown in FIG. 6, the tractor **100A** according to the present preferred embodiment includes a LiDAR sensor **110**, a GNSS unit **120**, and one or more obstacle sensors **130**. Although the example of FIG. 6 illustrates one obstacle sensor **130**, obstacle sensors **130** may be provided at a plurality of positions of the tractor **100A**.

(112) As shown in FIG. 7, the tractor **100A** includes a vehicle body **101**, a prime mover (engine) **102**, and a transmission **103**. On the vehicle body **101**, wheels (tires) **104** and a cabin **105** are provided. The wheels **104** include front wheels **104F** and rear wheels **104R**. Inside the cabin **105**, a driver's seat **107**, a steering device **106**, and an operation terminal **200** are provided. Either or both of the front wheels **104F** or the rear wheels **104R** may be crawlers, rather than tires.

(113) The LiDAR sensor **110** in the present preferred embodiment is placed in a lower portion of the front surface of the vehicle body **101**. The LiDAR sensor **110** may be placed at a position that is lower than an average height of the trunks of the rows of trees existing in an environment in which the tractor **100A** travels, e.g., at a height of not less than about 15 cm and not more than about 100 cm from the ground surface. The LiDAR sensor **110** may be placed at a position that is lower than a half of the height of the tractor **100A**, for example. The height of the tractor **100A** is a height from the ground surface to the topmost portion of the cabin **105**, e.g., about 2 meters or more. The LiDAR sensor **110** may be placed at a position that is lower than about $\frac{1}{3}$ of the height of the tractor **100A**. In the example of FIG. 7, the LiDAR sensor **110** is placed at a position that is lower than the tops of the wheels **104** and headlights **109**. Effects that are obtained by placing the LiDAR sensor **110** at such a low position will be described later.

(114) While the tractor **100A** is moving, the LiDAR sensor **110** repeatedly outputs sensor data indicating the distances and directions, or two-dimensional or three-dimensional coordinate values, of objects existing in the surrounding environment. The sensor data that is output from the LiDAR sensor **110** is processed by a controller, such as an ECU (Electronic Control Unit), that is included in the tractor **100A**. By using the aforementioned SLAM algorithm, the controller is able to perform processes such as generation of environment map data based on the sensor data, and localization using the environment map. Instead of completing the environment map data, the controller may generate some local map data from which to generate the environment map data. In that case, the process of integrating local map data to establish environment map data may be performed on a computer that is external to the tractor **100A**, e.g., a cloud server.

(115) In the example of FIG. 7, the obstacle sensor(s) **130** is provided at the rear of the vehicle body **101**. The obstacle sensor(s) **130** may be placed at any other position than the rear of the vehicle body **101**. For example, one or more obstacle sensors **130** may be provided at any position selected from among the sides of the vehicle body **101**, the front of the vehicle body **101**, and the cabin **105**. The obstacle sensor(s) **130** detects objects that are relatively close to the tractor **100A**. Each obstacle sensor **130** may include a laser scanner or an ultrasonic sonar, for example. When an object exists at a position closer to the obstacle sensor **130** than a predetermined distance, the obstacle sensor **130** outputs a signal indicating the presence of an obstacle. A plurality of obstacle sensors **130** may be provided at different position of the body of the tractor **100A**. For example, a plurality of laser scanners and a plurality of ultrasonic sonars may be placed at different positions of the body. Providing a multitude of obstacle sensors **130** can reduce blind spots in monitoring obstacles around the tractor **100A**. The LiDAR sensor **110** may function as an obstacle sensor.

(116) The GNSS unit **120** is disposed above the cabin **105**. The GNSS unit **120** is a GNSS receiver that includes an antenna to receive signals from a GNSS satellite and a processing circuit. The

GNSS unit **120** receives GNSS signals which are transmitted from a GNSS satellite, such as the GPS (Global Positioning System), GLONAS, Galileo, BeiDou, or QZSS (Quasi-Zenith Satellite System, e.g., MICHIBIKI), and performs positioning based on the signals. The tractor **100A** according to the present preferred embodiment is mainly used in environments where multiple trees grow to make it difficult to use a GNSS, e.g., a vineyard, which is the reason why the LiDAR sensor **110** is used in positioning. However, in an environment where it is possible to receive GNSS signals, positioning may be performed by using the GNSS unit **120**. By combining the positioning based on the LiDAR sensor **110** and the positioning based on the GNSS unit **120**, the stability or accuracy of positioning can be improved.

(117) The prime mover **102** may be a diesel engine, for example. Instead of a diesel engine, an electric motor may be used. The transmission **103** can switch the propulsion of the tractor **100A** through a speed changing mechanism. The transmission **103** can also switch between forward travel and backward travel of the tractor **100A**.

(118) The steering device **106** includes a steering wheel, a steering shaft connected to the steering wheel, and a power steering device to assist in the steering by the steering wheel. The power steering device includes a hydraulic device or an electric motor to supply an assisting force to rotate the steering shaft. The front wheels **104F** are the wheels responsible for steering, such that changing their steering angle can cause a change in the traveling direction of the tractor **100A**. The steering angle of the front wheels **104F** can be changed by manipulating the steering wheel. When automatic steering is performed, under the control of an electronic control unit (ECU) disposed in the tractor **100A**, the steering angle is automatically adjusted by the power of the hydraulic device or electric motor.

(119) A linkage device **108** is provided at the rear of the vehicle body **101**. The linkage device **108** may include, e.g., a three-point linkage (hitch) device, a PTO (Power Take Off) shaft, and a universal joint. The linkage device **108** allows the implement **300** to be attached to or detached from the tractor **100A**. The linkage device **108** is able to raise or lower the three-point linkage device with a hydraulic device, for example, thus controlling the position or attitude of the implement **300**. Moreover, motive power can be sent from the tractor **100A** to the implement **300** via the universal joint. While towing the implement **300**, the tractor **100A** allows the implement **300** to perform a predetermined task. The linkage device may be provided frontward of the vehicle body **101**. In that case, the implement may be connected frontward of the tractor **100A**. In the case where the LiDAR sensor **110** is used while the implement is connected frontward of the tractor **100A**, the LiDAR sensor **110** is to be placed in a position where the laser beam is not obstructed by the implement.

(120) Although the implement **300** shown in FIG. 7 is a sprayer that sprays chemical agents onto crops, the implement **300** is not limited to a sprayer. For example, any arbitrary implement such as a mower, a seeder, a spreader, a rake, a baler, a harvester, a plow, a harrow, or a rotary tiller may be connected to the tractor **100A** for use.

(121) In the following description, the uvw coordinate system shown in FIG. 7 is used. The uvw coordinate system is a coordinate system that is fixed to the tractor **100A** and the LiDAR sensor **110**. The traveling direction of the tractor **100A** is the u-axis direction, the left direction is the v-axis direction, and the upper direction is the w-axis direction. Although the origin of the uvw coordinate system is illustrated to be at a distant position from the tractor **100A** for ease of understanding in the figure, in practice, the origin of the uvw coordinate system is matched to the center of the LiDAR sensor **110** provided on the tractor **100A**.

(122) FIG. 8 is a block diagram showing an exemplary schematic configuration of the tractor **100A** and the implement **300**. In addition to the LiDAR sensor **110**, the GNSS unit **120**, the obstacle sensor(s) **130**, and the operation terminal **200**, the tractor **100A** in this example also includes an inertial measurement unit (IMU) **125**, a drive device **140**, a storage device **150**, a plurality of electronic control units (ECU) **160**, **170** and **180**, and a communication interface (IF) **190**. The

implement **300** includes a drive device **340**, a controller **380**, and a communication interface **390**. Note that FIG. **8** shows component elements which are relatively closely related to the localization operation, the self-driving operation, and the map generation operation according to the present preferred embodiment, while other component elements are omitted from illustration.

(123) The IMU **125** in the present preferred embodiment includes a 3-axis accelerometer and a 3-axis gyroscope. The IMU **125** functions as a motion sensor which can output signals representing parameters such as acceleration, velocity, displacement, and attitude of the tractor **100A**. The IMU **125** may output such signals as frequently as several tens to several thousands of times per second, for example. Instead of the IMU **125**, a 3-axis accelerometer and a 3-axis gyroscope may be separately provided.

(124) For example, the drive device **140** may include various devices that are needed for the traveling of the tractor **100A** and the driving of the implement **300**, e.g., the prime mover **102**, transmission **103**, wheels **104**, steering device **106**, and linkage device **108**. The prime mover **102** includes an internal combustion engine such as a diesel engine. Instead of an internal combustion engine or in addition to an internal combustion engine, the drive device **140** may include an electric motor that is dedicated to traction purposes.

(125) The storage device **150** includes one or more storage media such as a flash memory or a magnetic disc, and stores various data that are generated by the sensors and the ECUs **160**, **170** and **180**, such as environment map data. The storage device **150** also stores data such as a computer program(s) to cause the ECUs **160**, **170** and **180** to perform various operations to be described later. Such a computer program(s) may be provided for the tractor **100A** via a storage medium (e.g., a semiconductor memory or an optical disc) or through telecommunication lines (e.g., the Internet). Such a computer program(s) may be marketed as commercial software.

(126) The ECU **160** is an electronic circuit that performs processing based on the aforementioned SLAM technique. The ECU **160** includes a localization module **162** and a map data generating module **164**. The ECU **160** is an example of the aforementioned localization processor and data generator. Based on the data or signals that are repeatedly output from the LiDAR sensor **110** and the IMU **125** while the tractor **100A** is traveling, the ECU **160** can generate map data while estimating the position and orientation (i.e., the pose) of the tractor **100A**). Moreover, during the self-driving after generation of the environment map data, the ECU **160** performs matching between the sensor data that is output from the LiDAR sensor **110** and the environment map data, thus being able to estimate the position and orientation of the tractor **100A**.

(127) The localization module **162** performs computations to achieve estimation of the position and orientation of the tractor **100A**, i.e., localization. The map data generating module **164** performs the process of generating map data. The “map data” generated by the map data generating module **164** includes environment map data to be used for the matching during self-driving, and local map data, which is partial data that is generated in order to establish the environment map data. By joining together repeatedly generated local map data, the map data generating module **164** may generate the environment map data, and record it to the storage device **150**. The localization module **162** and the map data generating module **164** may be implemented by a single circuit, or divided into a plurality of circuits.

(128) At a stage where no environment map data has been generated yet, while performing localization, the ECU **160** may detect trunks of the rows of trees in the surrounding environment of the tractor **100A**, repeatedly generate local map data indicating a distribution of the detected trunks of the rows of trees, and record it to the storage device **150**. Furthermore, by joining together local map data, the ECU **160** may generate environment map data concerning the entire field (e.g., vineyard), or one section of the field. Environment map data may be generated for each section of the field. Without detecting trunks of the rows of trees at the stage of generating local map data, the ECU **160** may detect the trunks at the stage of generating final environment map data, and record the trunks in a format distinguishable from other objects. During the self-driving after generation of

the environment map data, the ECU **160** performs matching between the sensor data that is output from the LiDAR sensor **110** and the environment map data, thus estimating the position and orientation of the tractor **100A**. Note that the ECU **160** may determine only the position of the tractor **100A** through matching, and determine the orientation of the tractor **100A** by utilizing the signals from the IMU **125**.

(129) The ECU **170** is a circuit that performs the process of determining a path of the tractor **100A**. At a stage where no environment map data has been generated yet, the ECU **170** determines a path to be traveled by the tractor **100A** based on the data or signals output from the LiDAR sensor **110** and the obstacle sensor(s) **130**. For example, a path which goes between multiple rows of trees as detected based on the sensor data that is output from the LiDAR sensor **110** and which avoids obstacles is determined as a target path. During the self-driving after generation of the environment map data, based on the environment map data or on instructions from the user, the ECU **170** determines a target path (hereinafter also referred to as an “intended travel path”).

(130) The ECU **180** is a circuit that controls the drive device **140**. The ECU **180** controls the drive device **140** based on the position and orientation of the tractor **100A** estimated by the ECU **160**, and the intended travel path determined by the ECU **170**. The ECU **180** also performs the operation of generating a signal to control the operation of the implement **300**, and transmitting this signal from the communication I/F **190** to the implement **300**.

(131) The ECUs **160**, **170** and **180** may communicate with one another according to a vehicle bus standard such as CAN (Controller Area Network). Although the ECUs **160**, **170** and **180** are illustrated as individual blocks in FIG. **8**, each of these functions may be implemented by a plurality of ECUs. Alternatively, a single onboard computer that integrates the functions of the ECUs **160**, **170** and **180** may be provided. The tractor **100A** may include ECUs other than the ECUs **160**, **170** and **180**. Any number of ECUs may be provided in accordance with functionality.

(132) The communication I/F **190** is a circuit that performs communications with the communication I/F **390** of the implement **300** of the server **500**. The communication I/F **190** performs exchanges of signals complying with a communication control standard such as ISOBUS under ISO 11783, for example, between itself and the communication I/F **390** of the implement **300**. This causes the implement **300** to perform a desired operation, or allows information to be acquired from the implement **300**. Moreover, the communication I/F **190** can communicate with an external computer via a wired or wireless network. For example, it may communicate with a server or other computer in a farming management system that manages the growth status of crops, the operating status of the tractor **100A**, work records, and so on.

(133) The operation terminal **200** is a terminal for the user to perform a manipulation related to the self-driving or automatic steering of the tractor **100A**, and may be referred to as a virtual terminal (VT). The operation terminal **200** may include a display device such as a touch screen panel, and/or one or more buttons. By manipulating the operation terminal **200**, the user can perform various manipulations, such as switching ON/OFF the self-driving mode or the automatic steering mode, setting an initial position of the tractor **100A**, setting a path, recording or editing an environment map.

(134) The drive device **340** in the implement **300** performs a necessary operation for the implement **300** to perform a predetermined task. The drive device **340** includes devices adapted to the intended use of the implement **300**, e.g., a pump, a hydraulic device, or an electric motor.

(135) The controller **380** controls the operation of the drive device **340**. In response to a signal that is transmitted from the tractor **100A** via the communication I/F **390**, the controller **380** causes the drive device **340** to perform various operations.

(136) Next, with reference to FIGS. **9A** and **9B**, an exemplary configuration of the LiDAR sensor **110** will be described. The LiDAR sensor **110** in the present preferred embodiment is a scan-type sensor capable of acquiring information on the distance distribution of objects within a space by laser beam scanning. FIG. **9A** is a schematic diagram of the LiDAR sensor **110** as viewed in a

lateral direction of the tractor **100A**. FIG. **9B** is a schematic diagram of the LiDAR sensor **110** as viewed from vertically above. In FIGS. **9A** and **9B**, straight lines extending radially schematically represent the center axes (or traveling directions) of laser beams emitted from the LiDAR sensor **110**. Although each laser beam is collimated into parallel light, it has an angle of spread of several milli-radians (e.g., about 0.1 to about 0.2 degrees). Therefore, the cross-sectional size (spot diameter) of each laser beam increases in proportion to distance from the LiDAR sensor **110**. For example, a light spot with a diameter of several centimeters may be formed about 20 meters away from the LiDAR sensor **110**. In the figure, for simplicity, the spread of each laser beam is ignored, and only the center axis of the laser beam is illustrated.

(137) The LiDAR sensor **110** in the example of FIG. **9A** is able to emit laser beams from a plurality of laser light sources that are arranged along the vertical direction, respectively at different angles of elevation. An angle of elevation is defined as an angle relative to the uv plane. In this example, the uv plane is essentially parallel to the horizontal plane. Note that, when the ground surface is inclined with respect to the horizontal plane, the uv plane and the horizontal plane will intersect. FIG. **9A** illustrates N laser beams $L_{sub.1}, \dots, L_{sub.N}$ being emitted. Herein, “ N ” is an integer of 1 or greater, and may be e.g., 10 or greater, and 64, or even 100 or greater, for high-performance models. Among the plurality of laser beams, a $k_{sup.th}$ laser beam from the bottom has an angle of elevation $\theta_{sub.k}$. As one example, FIG. **9A** shows an angle of elevation $\theta_{sub.N-1}$ of an $N-1$ th laser beam. The angle of elevation of any laser beam going upward from the uv plane is defined as a “positive angle of elevation”, whereas the angle of elevation of any laser beam going downward from the uv plane is defined as a “negative angle of elevation”.

(138) A LiDAR sensor having an N of 1 may be referred to as a “two-dimensional LiDAR”, while a LiDAR sensor having an N of 2 or more may be referred to as a “three-dimensional LiDAR”. When N is 2 or more, the angle made by the first laser beam and an $N_{sup.th}$ laser beam is referred to as the “vertical viewing angle”. The vertical viewing angle may be set in a range from about 20° to about 60° , for example.

(139) As shown in FIG. **9B**, the LiDAR sensor **110** is able to change the outgoing directions (azimuth angles) of laser beams. FIG. **9B** shows the outgoing directions of the plurality of laser beams shown in FIG. **9A** as rotating around a rotation axis that is parallel to the w axis. The range of the outgoing directions (azimuth angles) of the laser beams may be about 360° , or an angle range that is smaller than about 360° (e.g., about 210° or about 270°). The range of azimuth angles of the outgoing directions of laser beams is referred to as the “horizontal viewing angle”. The horizontal viewing angle may be set in a range from about 90° to about 360° , for example. While rotating the outgoing directions of laser beams around a rotation axis that is parallel to the w axis, the LiDAR sensor **110** sequentially emits pulsed laser light (laser pulses) in directions of different azimuth angles. Thus, with pulsed laser light emitted at different angles of elevation and different azimuth angles, it is possible to measure the distance to each reflection point. Each reflection point corresponds to each individual point included in the point cloud data. The operation of measuring the distance to each reflection point while the azimuth angles of laser beams make one rotation around the rotation axis is referred to as one scan. Sensor data that is obtained through one scan includes data that is measured for every layer that is associated with a certain angle of elevation as shown in FIG. **9A**. Therefore, the greater the number of layers is, the greater the number of points in the point cloud that are obtained through one scan for the same environment. The LiDAR sensor **110** repeats scan operations with a frequency of about 1 to 20 times per second, for example. During one scan operation, 100,000 or more pulses of laser light may be emitted in different directions, for example.

(140) FIG. **10** is a block diagram showing an exemplary schematic configuration of the LiDAR sensor **110**. As shown in FIG. **10**, the LiDAR sensor **110** includes multiple laser units **111**, an electric motor **114**, a control circuit **115**, a signal processing circuit **116**, and a memory **117**. Each laser unit **111** includes a laser light source **112** and a photodetector **113**. Each laser unit **111** may

include optics such as a lens(es) and a mirror(s), but they are omitted from illustration. By rotating a mirror that is placed on the optical path of a laser beam emitted from each laser light source **112**, for example, the motor **114** changes the direction of the laser beam emitted from the respective laser light source **112**.

(141) Each laser light source **112** includes a laser diode, and emits a pulsed laser beam of a predetermined wavelength in response to a command from the control circuit **115**. The wavelength of the laser beam may be a wavelength that is included in the near-infrared wavelength region (approximately 700 nm to approximately 2.5 μm), for example. The wavelength used depends on the material of the photoelectric conversion element used for the photodetector **113**. In the case where silicon (Si) is used as the material of the photoelectric conversion element, for example, a wavelength around 900 nm may be mainly used. In the case where indium gallium arsenide (InGaAs) is used as the material of the photoelectric conversion element, a wavelength of not less than about 1000 nm and not more than about 1650 nm may be used, for example. Note that the wavelength of the laser beam is not limited to the near-infrared wavelength region. In applications where influences of ambient light are not a problem (e.g., for nighttime use), a wavelength included in the visible region (approximately 400 nm to approximately 700 nm) may be used. Depending on the application, the ultraviolet wavelength region may also be used. In the present specification, any radiation in the ultraviolet, visible light, and infrared wavelength regions in general is referred to as “light”.

(142) Each photodetector **113** is a device to detect laser pulses that are emitted from the laser light source **112** and reflected or scattered by an object. The photodetector **113** includes a photoelectric conversion element such as an avalanche photodiode (APD), for example. The photodetector **113** outputs an electrical signal which is in accordance with the amount of received light.

(143) In response to a command from the control circuit **115**, the motor **114** rotates the mirror that is placed on the optical path of a laser beam emitted from each laser light source **112**. This realizes a scan operation that changes the outgoing directions of laser beams.

(144) The control circuit **115** controls emission of laser pulses by the laser light sources **112**, detection of reflection pulses by the photodetectors **113**, and rotational operation by the motor **114**. The control circuit **115** can be implemented by a circuit that includes a processor, e.g., a microcontroller unit (MCU), for example.

(145) The signal processing circuit **116** is a circuit to perform computations based on signals that are output from the photodetectors **113**. The signal processing circuit **116** uses a ToF (Time of Flight) technique to calculate a distance to an object that has reflected a laser pulse emitted from a laser light source **112**, for example. ToF techniques include direct ToF and indirect ToF. Under direct ToF, the time from the emission of a laser pulse from the laser light source **112** until reflected light is received by the photodetector **113** is directly measured to calculate the distance to the reflection point. Under indirect ToF, a plurality of exposure periods are set in the photodetector **113**, and the distance to each reflection point is calculated based on a ratio of light amounts detected in the respective exposure periods. Either the direct ToF or indirect ToF method may be used. The signal processing circuit **116** generates and outputs sensor data indicating the distance to each reflection point and the direction of that reflection point, for example. Furthermore, the signal processing circuit **116** may calculate coordinates (u,v) or (u,v,w) in the sensor coordinate system based on the distance to each reflection point and the direction of that reflection point, and include these in the sensor data for output.

(146) Although the control circuit **115** and the signal processing circuit **116** are two separate circuits in the example of FIG. **10**, they may be implemented as a single circuit.

(147) The memory **117** is a storage medium to store data that is generated by the control circuit **115** and the signal processing circuit **116**. For example, the memory **117** stores data that associates the emission timing of a laser pulse emitted from each laser unit **111**, the outgoing direction, the reflected light intensity, the distance to the reflection point, and the coordinates (u,v) or (u,v,w) in

the sensor coordinate system. Such data is generated each time a laser pulse is emitted, and recorded to the memory **117**. The control circuit **115** outputs such data with a predetermined cycle (e.g., the length of time required to emit a predetermined number of pulses, a half scan period, or one scan period). The output data is recorded in the storage device **150** of the tractor **100A**.
(148) Note that the method of distance measurement is not limited to the ToF technique, but other methods such as the FMCW (Frequency Modulated Continuous Wave) technique may also be used. In the FMCW technique, light whose frequency is linearly changed is emitted, and distance is calculated based on the frequency of beats that occur due to interferences between the emitted light and the reflected light.

(149) Next, the operation of the tractor **100A** will be described.

(150) FIG. **11** is a diagram schematically showing an example of an environment in which the tractor **100A** travels. In this example, while traveling between multiple rows of trees **20** in a vineyard, the tractor **100A** uses the implement **300** to perform predetermined tasks (e.g., mowing, preventive pest control, or the like). The sky over a vineyard is obstructed by branches and leaves, thus hindering self-driving using a GNSS. Also in a vineyard, the leaves of trees or the outer shapes of hedges change significantly with seasons. This makes it difficult use a point cloud map that is generated by conventional SLAM techniques throughout the seasons.

(151) Therefore, in a preferred embodiment of the present disclosure, an environment map that is suitable for an orchard such as a vineyard is prepared, and self-driving of the tractor **100A** is performed based on this environment map. For example, once planted, trees in a vineyard are unlikely to be replanted for long periods of time, and their trunks are less susceptible to seasonal changes in outer shape than are their leaves. By using trunks as landmarks for SLAM, it is possible to generate an environment map that is usable throughout the seasons, and perform self-driving tasks without having to regenerate environment maps throughout the year.

(152) Hereinafter, operation of the tractor **100A** will be specifically described. By using the sensor data that is output from the LiDAR sensor **110**, the ECU **160** first generates environment map data indicating a distribution of trunks of rows of trees **20** in the entire vineyard or one section of the vineyard, and records it to the storage device **150**. Generation of the environment map data is performed by repeating localization by the localization module **162** and generation of local map data by the map data generating module **164**. The map data generating module **164** repeats the operation of, while performing localization based on sensor data that is repeatedly output from the LiDAR sensor **110** while the tractor **100A** is moving, detecting trunks of the rows of trees in the surrounding environment of the tractor **100A**, and generating local map data indicating a distribution of the detected trunks of the rows of trees and recording it to the storage device **150**. The map data generating module **164** generates the environment map data by joining together local map data that are generated while the tractor **100A** travels in the entire vineyard or one section of the vineyard. This environment map data is data recorded in a format that allows the distribution of trunks of the rows of trees **20** in the environment to be distinguished from other objects. Thus, in the present preferred embodiment, the trunks of the rows of trees **20** are used as landmarks for SLAM.

(153) One or more posts may be provided in an area where multiple rows of trees are placed. For example, in a vineyard, generally, multiple posts are provided near the trees to create a hedge construction. Similarly to the tree trunks, posts are less likely to change their outer shape from season to season, and therefore are suitable as landmarks. Therefore, the localization module **162** may further detect posts in the surrounding environment of the tractor **100A** based on sensor data, and the map data generating module **164** may generate environment map data that indicates not only the distribution of the trunks but also the distribution of the posts.

(154) Once environment map data is generated, self-driving or autonomous driving by the tractor **100A** becomes possible. During travel of the tractor **100A**, the localization module **162** of the ECU **160** detects the trunks of the rows of trees **20** in the surrounding environment based on the sensor

data that is repeatedly output from the LiDAR sensor **110**, and performs matching between the detected trunks of the rows of trees **20** and the environment map data, thus estimating the position of the tractor **100A**. If the environment map data includes distribution information of posts, the localization module **162** may perform matching between the trunks of the rows of trees and the posts as detected based on the sensor data and the environment map data, thus estimating the position of the tractor **100A**. The ECU **180** for drive control is a controller to control the movement of the tractor **100A** in accordance with the position of the tractor **100A** estimated by the ECU **160**. For example, when deviating from an intended travel path determined by the ECU **170**, the ECU **180** adjusts steering of the tractor **100A** so as to come closer to the intended travel path. Such steering control may be performed based not only on the position but also on the orientation of the tractor **100A**.

(155) An environment map indicating a distribution of trunks of the rows of trees **20** can be generated at any arbitrary timing. For example, the environment map may be generated in a season when there are few leaves on the trees, e.g., winter. In that case, it is easier to generate an environment map that more accurately reflects the distribution of the trunks than in the case of generating an environment map in a season with many leaves on the trees, e.g., summer. When the environment map is generated in winter, the ECU **160** may perform localization based on that environment map not only in winter but also in any other season. In that case, the ECU **160** performs localization by performing matching between the environment map data generated during winter and the data obtained by eliminating portions other than the trunks from the sensor data that is output from the LiDAR sensor **110**.

(156) The ECU **160** may generate data indicating a distribution of the trunks of the rows of trees as detected based on the sensor data that is repeatedly acquired during travel of the tractor **100A**, and update the environment map data by using this data. For example, the ECU **160** may update the environment map data by adding information of the trunks detected from newly acquired sensor data to the environment map data that was generated in the previous run. Alternatively, data indicating the distribution of the detected trunks of the rows of trees may be transmitted to an external device that updates the environment map data. In that case, the ECU **160** includes information indicating the estimated position of the tractor **100A** in the data indicating the distribution of the trunks (e.g., local map data) and outputs it. The external device may update the environment map data by using the acquired data, and transmit an updated environment map data to the tractor **100A**. Through such operation, even if the traveling environment has changed since the point in time of generating the environment map due to tree growth or the like, an environment map that reflects such change can be newly generated.

(157) FIG. **12** is a perspective view schematically showing an example of the surrounding environment of the tractor **100A**. The tractor **100A** travels between two adjacent rows of trees in a vineyard. While traveling, the tractor **100A** uses the LiDAR sensor **110** to scan the surrounding environment with laser beams. In this manner, data indicating a distance distribution of objects existing in the environment is acquired. The data indicating the distance distribution is converted into two-dimensional or three-dimensional point cloud data, and recorded. By integrating the point cloud data repeatedly acquired while traveling in the vineyard, the tractor **100A** generates environment map data indicating a distribution of the trunks of the rows of trees in the entire vineyard, or one section of the vineyard. Integration of the point cloud data includes the process of converting the coordinates of multiple reflection points acquired at different timings into coordinates in a world coordinate system that is fixed to the globe, and joining them together. In the present preferred embodiment, the ECU **160** of the tractor **100A** generates the environment map data. Alternatively, a computer provided externally to the tractor **100A** may generate the environment map data.

(158) FIG. **13A** is a diagram schematically showing an example of a travel path of the tractor **100A** when generating an environment map and during self-driving. The tractor **100A** travels between

multiple rows of trees **20** in the vineyard along a path **30** indicated by an arrow in FIG. **13A**, for example. Although this figure illustrates any line segment included in the path **30** to be a straight line, the path along which the tractor **100A** actually travels may include meandering portions. Now, the multiple rows of trees **20** are sequentially designated as a first row of trees **20A**, a second row of trees **20B**, a third row of trees **20C**, a fourth row of trees **20D**, . . . , from the end. In the example of FIG. **13A**, the tractor **100A** first travels between the first row of trees **20A** and the second row of trees **20B**, and upon completing this travel, turns around to travel between the second row of trees **20B** and the third row of trees **20C** in the opposite direction. Once the travel between the second row of trees **20B** and the third row of trees **20C** is completed, it further turns to travel between the third row of trees **20C** and the fourth row of trees **20D**. Thereafter, by repeating a similar operation, it travels to the final end of the path **30**, which will be in between the last two rows of trees. In the case where the distance between adjacent rows of trees is short, as shown in FIG. **13B**, it may travel so as to go along every other row. In this case, after the travel between the last two rows of trees is completed, an operation of traveling between untraveled rows of trees, so as to go along every other row, may be performed. Such travel is automatically performed as the tractor **100A** conducts localization by using the LiDAR sensor **110**. At timings when the GNSS unit **120** is able to receive a GNSS signal, positioning may be conducted based on the GNSS signal. For example, at any timing of turning around along the path **30** illustrated in either FIG. **13A** or **13B**, no leaves exist to obstruct the GNSS signal, and therefore positioning based on the GNSS signal is possible. (159) Next, with reference to FIG. **14A** to FIG. **17C**, an example of sensor data acquisition by the tractor **100A** will be described. When an environment map has already been established, sensor data is used to estimate the position of the tractor **100A** through matching against the environment map. On the other hand, when no environment map has been established, or at least a portion of an environment map is to be updated, establishment or update of the environment map is performed based on sensor data. The establishment and update of the environment map do not need to be made during travel of the tractor **100A**, but may be carried out as the sensor data acquired during travel of the tractor **100A** is processed inside the tractor **100A** or at an external computer.

(160) FIG. **14A** is a diagram schematically showing a tree **21** and the ground surface being irradiated with laser beams emitted from the LiDAR sensor **110** of the tractor **100A**. FIG. **14B** is a diagram schematically showing the ground surface being irradiated with laser beams emitted from the LiDAR sensor **110**. In these figures, laser beams which are emitted in directions of negative angles of elevation from the LiDAR sensor **110** are depicted with thick broken lines, whereas laser beams which are emitted in directions of positive angles of elevation from the LiDAR sensor **110** are depicted with thin broken lines. Among the plurality of laser beams which are emitted at different angles of elevation from the LiDAR sensor **110**, laser beams which are emitted in directions of negative angles of elevation may, unless striking the tree **21**, be reflected from the ground surface. On the other hand, unless striking the tree **21**, laser beams which are emitted in directions of positive angles of elevation do not create reflection points, and the reflection points when striking the tree **21** are likely to be located on the surface of the leaves and branches. As described earlier, any reflection points located on the surface of the leaves and branches are not suitable for matching against the environment map data. Therefore, in the present preferred embodiment, reflection at the trunks is utilized to perform localization and establishment of the environment map. Specifically, trunks may be detected based on reflection points of, among the plurality of laser beams emitted at different angles of elevation from the LiDAR sensor **110**, laser pulses emitted from laser light sources whose angle of elevation is included in a predetermined range. Moreover, trunks may be detected based on reflection points whose height is lower than the average height of trunks. In these cases, after trunk detection, reflection points that are located on the ground surface may be selectively eliminated from the point cloud in the sensor data.

(161) FIGS. **15A** and **15B** are plan views schematically showing examples of point clouds in sensor data acquired through one scan operation by the LiDAR sensor **110** of the tractor **100A**. FIG. **15A**

schematically shows an example of a tractor **100A** and a distribution of reflection points of laser pulses radiated from the LiDAR sensor **110** at a certain point in time. In this example, the ground surface is assumed to be a flat surface. A laser beam that is reflected from the ground surface creates a reflection point(s) on a circular arc on the ground surface. Such reflection points on the ground surface are created by laser beams which are emitted in directions of negative angles of elevation. In FIG. **15A**, reflection points are present on seven circular arcs of different radii (corresponding to seven layers). Each layer is created through a scan in which a laser beam that is emitted in a different negative angle of elevation rotates in the azimuth angle direction. While this scan is being performed, a laser beam scans may also be performed in the horizontal direction or in any direction of a positive angle of elevation. Note that reflection points that are on the same layer are created by a laser beam that is emitted from the same laser light source at the same angle of elevation, as the laser beam strikes the object surface in pulse form while rotating. FIG. **15B** schematically shows an example of a tractor **100A** and a distribution of reflection points of laser pulses radiated from the LiDAR sensor **110** at a point in time slightly later than the point in time of FIG. **15A**.

(162) Some of the laser pulses radiated from the LiDAR sensor **110** are reflected at the surface of trunks **22** of trees. Some of the laser pulses that are reflected from the ground surface, the trunks **22** of trees, or other objects are detected by the LiDAR sensor **110** and their distances to the reflection points are measured, unless the reflection points are located far beyond measurable distance (e.g., about 50 m, about 100 m, or about 200 m). For example, for each reflection point, the LiDAR sensor **110** generates sensor data that associates a distance to the reflection point, a direction of the reflection point, a reflected light intensity, and an identification number of the laser light source that emitted the laser beam that has created the reflection point. This sensor data may also include information of a time of measurement. The time of measurement information may be recorded for each reflection point, or a group of reflection points that were measured within a predetermined length of time, for example. The localization module **162** of the ECU **160** converts the sensor data that is output from the LiDAR sensor **110** into point cloud data. The point cloud data is data including information of three-dimensional coordinates (u,v,w) of each reflection point as expressed by a sensor coordinate system that is fixed to the LiDAR sensor **110**. In the case where the LiDAR sensor **110** converts the distance and direction data of each reflection point into point cloud data before outputting it, the localization module **162** omits conversion into point cloud data.

(163) When establishing an environment map, while the tractor **100A** is traveling, the map data generating module **164** generates, from the point cloud data, local map data in which the trunks **22** are recorded in a format that allows distinction from other portions. During travel of the tractor **100A**, the map data generating module **164** repeats the operation of adding local map data based on newly-acquired sensor data to already generated local map data, thus updating it. In this manner, the final environment map data can be generated. Note that the map data generating module **164** may only perform an operation of generating local map data and recording it during travel of the tractor **100A**, and perform generation of the final environment map data after completion of travel. In that case, the final environment map data may be generated by an external computer.

(164) In the present preferred embodiment, the LiDAR sensor **110** is attached to the front of the body of the tractor **100A**. Therefore, as shown in FIG. **15A** and FIG. **15B**, laser pulses are radiated frontward from the tractor **100A**. A LiDAR sensor **110** may also be attached to the rear of the tractor **100A**. In that case, point cloud data both frontward and rearward of the tractor **100A** can be acquired at one time, such that map data of a higher density of point cloud can be generated. Moreover, the accuracy of the matching-based localization during the self-driving after environment map generation can be improved. Furthermore, because point cloud data can be obtained from a broad range on the surface of each individual trunk **22**, the thickness of the trunk **22** can be measured with a high accuracy from the point cloud data thereof. Thus, the ECU **160** also functions as a calculator that measures the thickness of each detected trunk in the rows of trees

based on the sensor data, and records the thickness of each trunk to a storage. The storage may be a storage medium in the storage device **150**, or any other storage medium. The trunk thickness may be added to the environment map data as one piece of attribute information concerning each landmark, or recorded in any other map data of vector format.

(165) FIG. **16** is a diagram schematically showing an example point cloud of trunks **22** obtained through one scan of laser beams in multiple layers (L.sub.1, . . . , L.sub.k+3). FIGS. **17A** to **17C** are diagrams schematically showing a point cloud of trunks **22** being obtained by a scan of laser beams in three layers (L.sub.k, L.sub.k+1, L.sub.k+2), respectively, among the multiple layers shown in FIG. **16**. As shown in this figure, point cloud data indicating a distribution of trunks **22** existing in the measurable range can be obtained through every scan. More specifically, when the distance to each individual reflection point is measured through one scan, each layer may include regions in which reflection points of relatively long distance are clustered and regions in which reflection points of relatively short distance locally appear.

(166) FIG. **18** is a graph showing an example relationship between distances to reflection points that are measured by laser beams in multiple layers (L.sub.k, L.sub.k+1, L.sub.k+2, . . .) through one scan, and azimuth angles of the reflection points. The range of azimuth angles shown in FIG. **18** is so narrow that reflection from two trunks is shown in this range. In this example, main reflection points in each layer are located on the ground surface. Among the reflection points belonging to each individual layer, reflection points of relatively short distances are on a curve corresponding to the surface of a trunk. Regarding the two trunks, the trunk that is located at the shorter distance is irradiated with laser beams of layers L.sub.k, L.sub.k+1, L.sub.k+2, On the other hand, the trunk that is located at the longer distance is irradiated with laser beams of few layers L.sub.k+1, L.sub.k+2, Therefore, when the point cloud on the surface of a trunk in a three-dimensional space is projected onto a plane that is parallel to the horizontal plane, the point cloud on the trunk at the shorter distance have a higher density. Based on the measurement result shown in FIG. **18**, trunks can be detected. In reality, undulations or weeds may exist on the ground surface, and therefore measured values of distance are not as straightforward as illustrated in the figure.

(167) As the tractor **100A** repeats the aforementioned scan while moving, point cloud data with a high density can be acquired from each trunk located in the surroundings of the tractor **100A**. Therefore, by using large amounts of sensor data that are acquired by the moving tractor **100A**, it is possible to acquire point cloud data with a high density that is needed for the establishment of a high-accuracy environment map indicating a distribution of the trunks.

(168) Note that, instead of distances to reflection points, relationships between the reflected light intensity and the azimuth angles of reflection points can be used in detecting the trunks. The reason is that the intensity of reflected light associated with laser beams in the same layer increases as the distance to the reflection point decreases.

(169) As described earlier, sensor data (hereinafter also referred to as “scan data”) to be compared against the environment map indicating a distribution of the trunks may include many reflection points other than the reflection points that are located on the surface of trunks. Therefore, it is effective to select reflection points which are highly likely to be located on the surface of trunks from within the acquired scan data. Moreover, the scan data may produce point cloud data with a high density by integrating not only reflection points obtained through one scan but also reflection points obtained through multiple scans and acquired consecutively.

(170) Although omitted in FIGS. **17A** to **17C**, a laser beam that is emitted at a positive angle of elevation, for example, may create many reflection points because of irradiating leaves and branches other than the trunk. From the point cloud data generated based on scan data, the ECU **160** in the present preferred embodiment extracts some points possessing characteristic aspects of the trunk **22**, thus generating data indicating a distribution of trunks **22**.

(171) FIG. **19** shows an example of a local map that was generated by extracting reflection points

that are located on the surface of trunks from scan data that is acquired by the tractor **100A** being at a certain position. In this example, a two-dimensional local map indicating a two-dimensional distribution of trunks is generated. In such a local map, coordinates of any trunk that is located in the surroundings of the tractor **100A** at that position can be expressed by the uv coordinate system being fixed to the tractor **100A**.

(172) FIGS. **20A** and **20B** are plan views schematically showing a process of estimating the position of the tractor **100A** by matching the local map of FIG. **19** to an already-established environment map. The respective coordinates in the local map are expressed by the sensor coordinate system, whereas the respective coordinates in the environment map are expressed by the world coordinate system. Matching means optimizing the coordinate transformation from the sensor coordinate system to the world coordinate system. Through such optimization of coordinate transformation, the position coordinates and orientation of the sensor in the world coordinate system are determined. In FIGS. **20A** and **20B**, the trunks in the environment map are depicted in dark color. FIG. **20A** shows a state in the middle of matching. In this state, misalignments (errors) exist between the trunks on the local map and their corresponding trunks on the environment map. FIG. **20B** shows state where matching has been completed. In this state, the misalignments (errors) between the trunks on the local map and their corresponding trunks on the environment map are reduced or minimized.

(173) By thus using scan data in which the trunks are extracted and the environment map indicating a distribution of the trunks, localization with a high accuracy is achieved. Although the above example illustrates a matching based on a two-dimensional environment map, matching may be performed based on a three-dimensional environment map.

(174) Next, with reference to FIGS. **21A** and **21B**, example placements of the LiDAR sensor **110** will be described.

(175) FIG. **21A** shows an example where the LiDAR sensor **110** is attached to the lower front of the vehicle body of the tractor **100A**, as in the present preferred embodiment. FIG. **21B** shows an example where the LiDAR sensor **110** is attached to the upper front of the cabin of the tractor **100A**. In these figures, thick broken lines represent laser beams which are radiated against the trunk of a tree **21**, whereas thin broken lines represent laser beams which are emitted toward the leaves of the tree or the sky above them.

(176) In the example of FIG. **21A**, the LiDAR sensor **110** is mounted at a position that is lower than an average height of the trunks of the rows of trees. The average height of the trunks of the rows of trees is a mean value of the heights of the trunks of the trees included among the multiple rows of trees existing in an intended travel path of the tractor **100A**. The height of the trunk of a tree is the distance from the ground surface to the portion of the tree that produces the lowest branches. The LiDAR sensor **110** may be disposed at a height of e.g., not less than about 10 cm and not more than about 150 cm, and in one example not less than about 15 cm and not more than about 100 cm, from the ground surface. By placing the LiDAR sensor **110** at such a low position, it becomes possible to irradiate the trunk with more laser beams, without being obstructed by the leaves and branches. Therefore, it becomes possible to determine the position of the trunk from the sensor data with a high accuracy.

(177) On the other hand, in the example of FIG. **21B**, the LiDAR sensor **110** is placed at a high position (e.g., at a position of about 2 m from the ground surface), so that most of the laser beams irradiate not the trunk but the leaves. Therefore, the placement in FIG. **21B** is not suitable for applications where point cloud data of the trunk is to be acquired. However, such a placement can efficiently acquire distribution information of the leaves of the rows of trees, and therefore is effective in applications where distribution information of the leaves is to be actively acquired. For example, the placement of FIG. **21B** may be adopted for the purposes of managing the growth status of trees, or determining a path for avoiding the tree leaves. Even in the placement of FIG. **21B**, attaching the LiDAR sensor **110** in a downward orientation will allow the trunk to be irradiate

with more laser beams. Therefore, the placement shown in FIG. 21B may be adopted also in applications where a distribution data of the trunks is to be acquired. Without being limited to the lower front of the body or the upper front of the cabin, the LiDAR sensor **110** may be placed in other positions. For example, the LiDAR sensor **110** may be placed at a position lower than the height of the cabin.

(178) Although a single LiDAR sensor **110** is used in the present preferred embodiment, the tractor **100A** may include a plurality of LiDAR sensors. By combining sensor data that is output from the plurality of LiDAR sensors, distribution data of the trunks can be acquired more efficiently. The plurality of LiDAR sensors may be provided on the right and left of the tractor **100A**, or at the front and the rear, for example. Details of preferred embodiments of mobile machines including a plurality of LiDAR sensors will be described later.

(179) Among the plurality of laser light sources in the LiDAR sensor **110**, the localization module **162** in the ECU **160** may detect trunks based on reflection points of those laser pulses emitted from laser light sources whose angle of elevation is included in a predetermined range. For example, trunks may be detected based only on the reflection points of laser pulses which are emitted in directions of negative angles of elevation. The user may be allowed to set a range of angles of elevation of laser pulses to be used in trunk detection. For example, the user may be allowed to set a range of angles of elevation of laser pulses to be used for trunk detection through manipulation of the operation terminal **200**. Among the laser pulses emitted from the LiDAR sensor **110**, using only those laser pulses emitted in a specific range of angles of elevation which are highly likely to irradiate the trunks will allow the trunks to be detected more efficiently. Moreover, in the case where the ground surface has undulations so that the uv plane in the sensor coordinate system is significantly tilted from the horizontal plane, the range of angles of elevation of the laser beams may be adaptively selected in accordance with the angle of tilt of the sensor coordinate system, so that laser pulses reflected by the trunks are appropriately selected in extracting the reflection points. The angle of tilt of the LiDAR sensor **110** can be determined by utilizing signals from the IMU **125**.

(180) Next, an example of self-driving operation after the environment map is generated will be described.

(181) FIG. 22 is a diagram showing an exemplary functional configuration of the localization module **162** in the ECU **160**. By executing a computer program (software) stored in the storage device **150**, the localization module **162** performs matching between the scan data output from the LiDAR sensor **110** and the map data, thus estimating the position and orientation of the tractor **100A**.

(182) The localization module **162** performs the respective processes of scan data acquisition **162a**, map data acquisition **162b**, IMU data acquisition **162c**, scan data filtering **162d**, matching **162e**, and vehicle position/orientation determination **162f**. Hereinafter, details of these processes will be described.

(183) The localization module **162** acquires scan data that is output from the LiDAR sensor **110**. The LiDAR sensor **110** outputs scan data with a frequency of about 1 to 20 times per second, for example. This scan data may include the coordinates of multiple points expressed by the sensor coordinate system, and time stamp information. In the case where the scan data includes the information of distance and direction toward each point and not coordinate information, the localization module **162** performs conversion from the distance and direction information into coordinate information.

(184) The localization module **162** acquires the environment map data that is stored in the storage device **150**. The environment map data indicates a distribution of the trunks of the rows of trees included in the environment in which the tractor **100A** travels. The environment map data includes data in either one of formats (1) to (3) below, for example.

(185) (1) Data Recorded in a Format that Allows Trunks to be Distinguished from Objects Other

than Trunks

(186) For example, this may be data in which numerical value “1” is assigned to any point that is determined as a trunk, and numerical value “0” is assigned to any point that is determined as an object other than a trunk. A trunk ID for distinguishing each individual trunk may be included in the environment map data.

(187) (2) Data in which a Relatively Large Weight is Assigned to a Trunk, and a Relatively Small Weight is Assigned to any Object Other than a trunk

(188) For example, this may be data in which greater numerical values are assigned to points having a higher probability of being estimated as a point on the surface of a trunk.

(189) (3) Data Including Information of a Distribution of Detected Trunks, but not Including Information of a Distribution of Some or all of Objects Other than Trunks

(190) For example, this may be data resulting by eliminating all points but the points determined as the surface of trunks from the point cloud representing the environment map. Rather than eliminating all points that were determined not to be trunks, some points may be left. For example, in a vineyard, generally, posts are provided near the trunks to create a hedge construction. Information of points representing such posts may be included in the environment map.

(191) The determination as to whether or not a point in the point cloud corresponds to a trunk or post may be made based on whether a distribution of that point as well as multiple points surrounding that point is a distribution that reflects the surface shape of a trunk or a post (e.g., a distribution of a circular arc that projects downward), for example. Alternatively, from the data of point cloud in a curved distribution that is acquired through each scan, a collection of points whose distance from the LiDAR sensor **110** is locally shorter than those of neighboring points may be extracted, and these points may be determined as points representing trunks or posts. Data of a point cloud in a curved distribution that is acquired through each scan may be classified into a plurality of classes depending on distance from the LiDAR sensor **110**, and each class may be subjected to a determination as to whether it corresponds to a trunk or post or not. Moreover, a point cloud may be classified based not only on distance but also on reflection intensity information. Because the reflection intensity clearly differs between tree trunks and any other portions in the surroundings, it is effective to classify the point cloud based on similarity in reflection intensity and similarity in position. For example, multiple points whose reflection intensity is within a predetermined range and whose positions are close to one another may be regarded as prospective points representing the surface of a trunk. Laser beams of a plurality of different wavelengths may be emitted from the LiDAR sensor **110**, and a point cloud may be classified based on the ratio of reflection intensities for different wavelengths to detect trunks.

(192) Machine learning may be utilized for trunk detection. By utilizing a neural network-based machine learning algorithm such as deep learning, it becomes possible to detect points corresponding to the surface of trunks of trees from point cloud data with a high accuracy. In the case where a machine learning algorithm is used, generation of a trained model (i.e., learning) to detect trunks from the point cloud data is performed in advance.

(193) Not only tree trunks, but other objects may also be recognized. For example, through pattern recognition or machine learning, the ground surface, weeds, tree leaves, or the like may be recognized from the point cloud data and these points may be eliminated, thus generating point cloud data that mainly includes points corresponding to tree trunks. Prior to trunk detection, a process of extracting only a point cloud whose height from the ground surface is included within a predetermined range (e.g., about 0.5 m to about 1.5 m) may be performed. By regarding only a point cloud that is included in such a specific coordinate range as a target of trunk detection, the time required for detection can be reduced. Height from the ground surface is calculated by subtracting the Z coordinate of the ground surface from the Z coordinate of each point. The Z coordinate of the ground surface may be determined by referring to a digital elevation model (DEM), for example. The Z coordinate of the ground surface may be determined from a point cloud

representing the ground surface.

(194) FIG. **23** is a diagram schematically showing an example format of the environment map data. In this example, the environment map data includes information of the number of points, as well as information of an ID, a classification, a trunk ID, an X coordinate, a Y coordinate, a Z coordinate, and a reflection intensity of each point. Given a number n of points, n records of data are to be made, each record including a point ID, a classification, a trunk ID, an X coordinate, a Y coordinate, a Z coordinate, and a reflection intensity. The classification indicates what the point represents, and may include an identifier indicating tree trunks, hedge posts, the ground surface, weeds, tree leaves, or the like, for example. The classification may be a binary numerical value indicating whether it is a tree trunk or not, or a numerical value indicating the probability of being a tree trunk. The trunk ID is a number which, in the case where the point represents a trunk, distinguishes that trunk. The trunk ID may not be included in the environment map data. In addition to the trunk ID, identifiers of other objects (e.g., leaves, branches, the ground surface, and weeds) may be included. The X coordinate, the Y coordinate, and the Z coordinate are coordinates of that point in a world coordinate system. The reflection intensity represents an intensity of reflected light from that point. The reflection intensity information may not be included in the environment map data. Other than the information shown in FIG. **23**, the environment map data may include information of a header that indicates the data format and the like. In the case where the vineyard to be traveled by the tractor **100A** is broad, environment map data may be generated for each section of the vineyard (e.g., for each 100 m×100 m section). In that case, the environment map data may also include ID information for distinguishing sections. Note that, instead of recording three-dimensional coordinates of each point, the environment map data may record two-dimensional coordinates. For example, data including the coordinates of a point cloud on a plane which is at a specific height and which is parallel to the horizontal plane may be generated as environment map data. In that case, matching between the scan data and the environment map data is performed with respect to those two-dimensional coordinates.

(195) The localization module **162** acquires IMU data that is output from the IMU **125**. The IMU data may include information of the acceleration, velocity, displacement, attitude, time of measurement (time stamp), etc., of the tractor **100A**. The IMU data is output at a frequency of about several ten to several thousand times per second, for example. This output cycle is generally shorter than the output cycle of scan data by the LiDAR sensor **110**.

(196) FIG. **24** is a diagram showing an example temporal relationship between IMU data and scan data. In this example, the IMU data is output with a cycle of about 1 millisecond (ms), whereas the scan data is output with a cycle of about 10 ms. Given that the time required for one scan is e.g., about 100 ms, the scan data is output for every approximately 1/10 scan in the example of FIG. **24**. The scan data includes information of the distance and direction, or the coordinates, of each measured point as well as the time of measurement (time stamp).

(197) The localization module **162** refers to the time stamp of acquired scan data, and acquires the IMU data that was generated in the corresponding duration.

(198) The localization module **162** filters the acquired scan data so as to reduce the number of points to be subjected to matching. Furthermore, portions that are unnecessary for the matching are eliminated. For example, points that are determined as not corresponding to tree trunks or hedge posts, e.g., the ground surface, weeds, tree leaves, and obstacles, can be eliminated.

(199) The localization module **162** performs matching between the filtered scan data and the map data. The matching may be performed by using any arbitrary matching algorithm such as NDT (Normal Distribution Transform) or ICP (Iterative Closest Point), for example. Through the matching, the position and orientation of the LiDAR sensor **110** are determined.

(200) Based on the matching result, the localization module **162** determines the position and orientation of the tractor **100A**, and outputs data indicating the position and orientation. The data is sent to the ECU **180** for drive control, so as to be used in the control of the drive device **140**.

(201) Next, with reference to FIG. 25, the localization operation according to the present preferred embodiment will be described in more detail.

(202) FIG. 25 is a flowchart showing the localization operation by the localization module 162. By performing the operations from steps S101 to S109 shown in FIG. 25, the localization module 162 estimates the position and orientation, i.e., the pose, of the tractor 100A. Hereinafter, the operation of each step will be described. The operations shown in FIG. 25 may be begun by the user's manipulation using the operation terminal 200, for example.

(203) Step S101

(204) The localization module 162 reads environment map data from the storage device 150. In the case where different environment map data is recorded for each section of the vineyard to be traveled, the environment map data corresponding to the current point is read. The environment map data corresponding to the current point may be designated by the user manipulating the operation terminal 200, for example. Alternatively, in the case where a GNSS signal can be received at that point, the current point may be identified based on the GNSS signal received by the GNSS unit 120, and the corresponding environment map data may be selected and read. As is illustrated by this example, the processing can be performed rapidly by reading only partial environment map data that corresponds to the position of the tractor 100A. In this step, the entire environment map data may be read all at once.

(205) Step S102

(206) The localization module 162 sets a starting point of localization. The starting point of localization is the current position of the tractor 100A at the given point in time, which may be set by the user designating a specific point from a map which is displayed on the operation terminal 200, for example. Alternatively, in the case where a GNSS signal can be received at that point, a starting point may be set based on the GNSS signal received by the GNSS unit 120.

(207) Step S103

(208) Once the start position is set, the operation of the LiDAR sensor 110 is begun. The localization module 162 reads scan data that is output from the LiDAR sensor 110. The LiDAR sensor 110 outputs scan data with a predetermined cycle (e.g., not less than about 5 milliseconds and not more than about 1 second). The scan data may include, for each layer, point cloud data in the range of a few degrees to 360 degrees, for example. Every time scan data is output from the LiDAR sensor 110, for example, the localization module 162 reads that scan data. Alternatively, every time a predetermined number of instances of scan data are output, the localization module 162 may read such scan data altogether.

(209) Step S104

(210) The localization module 162 refers to the time stamp included in the scan data, and reads IMU data that corresponds to the scan data.

(211) Step S105

(212) Based on the IMU data having been read, the localization module 162 sets an initial position of matching. The initial position of matching is an estimated position of the tractor 100A at the current point in time indicated by the IMU data. By beginning matching from this initial position, the time until convergence can be reduced. Instead of using the IMU data, for example, an initial value of matching may be determined through linear interpolation, based on a difference between the estimated values of position and orientation from two scans in the past.

(213) Step S106

(214) The localization module 162 filters the acquired scan data to reduce the number of points used for matching. Furthermore, with the above-described method, trunks are detected, and at least a portion of the unwanted portions other than the trunks is eliminated from the point cloud.

(215) Now, with reference to FIG. 26, an example operation of scan data filtering will be described. FIG. 26 schematically shows a portion of the point cloud included in the scan data. In FIG. 26, white circles are the points included in the point cloud. The localization module 162 divides a

three-dimensional space in the sensor coordinate system into a plurality of voxels of constant size. Then, if multiple points are included in each individual voxel, such points are replaced by one point that is located at the centroid. In FIG. 26, the replaced points at centroids are expressed by dark circles. Through such downsampling, the number of points in the scan data can be reduced, thus making the matching rapid. In the present preferred embodiment, trunk detection is performed after such downsampling is performed. However, trunk detection may be performed before the downsampling. In the case where the data size of the point cloud is not an issue, the downsampling process may be omitted.

(216) Step S107

(217) The localization module 162 performs matching between the filtered scan data and the environment map data, thus estimating the pose of the LiDAR sensor 110. Specifically, by determining the coordinate transformation from the sensor coordinate system to the world coordinate system with a method such as NDT technique or ICP technique, the pose of the LiDAR sensor 110 is determined.

(218) Step S108

(219) Based on the pose of the LiDAR sensor 110, the localization module 162 calculates the pose of the tractor 100A, and outputs the result. The pose of the tractor 100A may be data of the coordinates (x,y,z) and attitude ($\theta_{sub.R}$, $\theta_{sub.P}$, $\theta_{sub.Y}$) of a representative point (origin) of the tractor 100A expressed by the world coordinate system. In the case where the environment map and the scan data represent a point cloud in a two-dimensional space (plane) and matching is to be performed in this two-dimensional space, the pose data to be output may include values of the two-dimensional coordinates (x,y) and orientation (θ). Only the position may be estimated through matching, while the attitude information indicated by the IMU data may straightforwardly be utilized as the attitude. In the case where the coordinate system of the tractor 100A matches the sensor coordinate system, step S108 may be omitted.

(220) Step S109

(221) The localization module 162 determines whether a command to end operation has been issued or not. A command to end operation may be issued when the user uses the operation terminal 200 to instruct that the self-driving mode be stopped, or when the tractor 100A has arrived at a destination, for example. If a command to end operation has not been issued, control returns to step S103, and a similar operation is performed with respect to the next scan data. If a command to end operation has been issued, the process is ended.

(222) Next, an example operation of the ECU 180 for drive control will be described.

(223) FIG. 27 is a flowchart showing an example operation of the ECU 180 to be performed after the pose of the tractor 100A has been estimated. At step S108 shown in FIG. 25, if a pose of the tractor 100A is output, the ECU 180 acquires data of the pose (step S121). Next, the ECU 180 calculates a deviation between the position of the tractor 100A indicated by the pose data and the intended travel path previously determined by the ECU 170 (step S122). The intended travel path may be set by the user manipulating the operation terminal 200, for example. The deviation represents a distance between the estimated position of the tractor 100A at the given point in time and the intended travel path. The ECU 180 determines whether the calculated deviation in position exceeds a pre-designated threshold value or not (step S123). If the deviation exceeds the threshold value, the ECU 180 changes the driving value (e.g., steering angle) of the drive device 140 so that the deviation decreases. In addition to the steering angle, the velocity may also be changed. If the deviation does not exceed the threshold value at step S123, step S124 is omitted. At the next step S125, the ECU 180 determines whether a command to end operation has been received or not. As mentioned above, a command to end operation may be issued when the user uses the operation terminal 200 to instruct that the self-driving mode be stopped, or when the tractor 100A has arrived at a destination, for example. If a command to end operation has not been issued, control returns to step S121, and a similar operation is performed on the basis of the pose of the tractor 100A as

estimated based on the next scan data. If a command to end operation has been issued, the process is ended.

(224) In the example of FIG. 27, the ECU 180 controls the drive device 140 based only on the deviation between the estimated position and the intended travel path, but the deviation in direction may further be taken into consideration in the control. For example, if a directional deviation, i.e., an angle difference between the estimated orientation of the tractor 100A as indicated by the acquired pose data and the direction of the intended travel path, exceeds a pre-designated threshold value, the ECU 180 may change the driving value (e.g., steering angle) of the drive device 140 in accordance with the deviation.

(225) Hereinafter, with reference to FIGS. 28A to 28D, an example of the steering control by the ECU 180 will be described.

(226) FIG. 28A is a diagram showing an example of a tractor 100A traveling along an intended travel path P. FIG. 28B is a diagram showing an example of a tractor 100A at a position which is shifted rightward from the intended travel path P. FIG. 28C is a diagram showing an example of a tractor 100A at a position which is shifted leftward from the intended travel path P. FIG. 28D is a diagram showing an example of a tractor 100A which is oriented in an inclined direction with respect to the intended travel path P.

(227) As shown in FIG. 28A, in the case where the position and orientation of the tractor 100A are not deviated from the intended travel path P, the ECU 180 maintains the steering angle and velocity of the tractor 100A without changing them.

(228) As shown in FIG. 28B, when the position of the tractor 100A is shifted rightward from the intended travel path P, the ECU 180 changes the steering angle by changing the angle of rotation of the steering motor included in the drive device 140 so that the traveling direction of the tractor 100A will be inclined leftward. Herein, not only the steering angle but also the velocity may be changed. The amount of change of the steering angle may be adjusted in accordance with the magnitude of a positional deviation Δx , for example.

(229) As shown in FIG. 28C, when the position of the tractor 100A is shifted leftward from the intended travel path P, the ECU 180 changes the steering angle by changing the angle of rotation of the steering motor so that the traveling direction of the tractor 100A will be inclined rightward. In this case, too, not only the steering angle but also the velocity may be changed. The amount of change of the steering angle may be adjusted in accordance with the magnitude of a positional deviation Δx , for example. As shown in FIG. 28D, in the case where the position of the tractor 100A is not considerably deviated from the intended travel path P but its orientation is nonetheless different from the direction of the intended travel path P, the ECU 180 changes the steering angle so that the directional deviation $\Delta\theta$ will become smaller. In this case, too, not only the steering angle but also the velocity may be changed. The amount of change of the steering angle may be adjusted in accordance with the magnitudes of the positional deviation Δx and directional deviation $\Delta\theta$, for example. For instance, the amount of change of the steering angle (which is in accordance with the directional deviation $\Delta\theta$) may be increased as the absolute value of the positional deviation Δx decreases. When the positional deviation Δx has a large absolute value, the steering angle will be changed greatly for returning to the intended travel path P, so that the directional deviation $\Delta\theta$ will inevitably have a large absolute value. Conversely, when the positional deviation Δx has a small absolute value, the directional deviation $\Delta\theta$ needs to become closer to zero, and therefore it makes sense to introduce a relatively large weight for the directional deviation $\Delta\theta$ in determining the amount of change of the steering angle.

(230) For the steering control and velocity control of the tractor 100A, control techniques such as PID control or MPC control (Model Predictive Control) may be applied. Applying these control techniques will make for smoothness of the control of bringing the tractor 100A closer to the intended travel path P.

(231) Note that, when an obstacle is detected by one or more obstacle sensors 130 during travel, the

ECU **180** controls the drive device **140** so as to avoid the obstacle. If the obstacle cannot be avoided, the ECU **180** halts the tractor **100A**. Note that, regardless of whether the obstacle is avoidable or not, the ECU **180** may halt the tractor **100A** whenever an obstacle is detected.

(232) Next, an example method of determining an intended travel path of the tractor **100A** by the ECU **170** will be described.

(233) After the environment map is generated, the ECU **170** determines an intended travel path of the tractor **100A**. The intended travel path may be automatically determined by the ECU **170** based on the environment map, or set by the user manipulating the operation terminal **200**.

(234) FIG. **29A** is a diagram showing an example of two-dimensional data indicating a trunk distribution of rows of trees that is generated based on the environment map. In addition to the environment map for matching purposes, the map data generating module **164** in the ECU **160** may generate two-dimensional data that roughly indicates a distribution of trunks **22** as shown in FIG. **29A**. This two-dimensional data expresses the trunk **22** of each individual tree with a circle of a constant size. Such two-dimensional data indicating a trunk distribution may be referred to as “trunk distribution data”. The trunk distribution data includes the information of position coordinates (x,y) of the center of each individual trunk, for example. In addition to the position coordinates, the trunk distribution data may include information of the identification number (trunk ID), trunk thickness, etc., of each individual trunk. Inclusion of such information allows for management of the growth status of the trunks. The growth status of the trunks or leaves, etc., of trees may be managed by a computer within a farming management system, for example. As is illustrated by this example, the growth status of trees may be managed by data which is distinct from the environment map. By using a human-machine interface (HMI) such as the operation terminal **200**, the user may be allowed to input information indicating the placement relationship of the multiple rows of trees, e.g., the intra-tree distance or the position coordinates of the trunks. The input information is recorded to the storage device **150** or another storage medium.

(235) FIG. **29B** is a diagram for describing an example method of determining an intended travel path **30** based on trunk distribution data. The ECU **170** in this example determines curves connecting the trunks **22** of the rows of trees, and determines curves or polylines that pass through the midpoint between every two adjacent rows of trees as the intended travel path **30**. It is not necessary for the center of the tractor **100A** to be located at the midpoint between two adjacent trees. Instead, the intended travel path **30** may be shifted from the midpoint between two adjacent trees. With reference to the determined intended travel path **30**, the ECU **180** for drive control performs steering control of the tractor **100A**.

(236) FIG. **30** is a diagram showing another example of trunk distribution data for path setting purposes. In this example, the traveling environment of the tractor **100A** is managed by being divided into a plurality of cells. Each cell may be a square, each of whose sides has a length that is close to a mean value of the diameters of the trunks **22**, for example. The cells are classified into the following four kinds (1) to (4): (1) a cell including a trunk **22**; (2) a cell that is adjacent to a cell including a trunk **22**; (3) a cell located between two adjacent rows of trees and not qualifying as (1) or (2); and (4) other cells.

(237) In FIG. **30**, cells (1) to (4) are expressed in different shades. The range of each cell (2) that is adjacent to a cell including a trunk **22** may be determined based on point cloud data indicating the leaves of the rows of trees that is acquired by the LiDAR sensor **110**. In that case, in addition to the environment map data for matching purposes, the map data generating module **164** of the ECU **160** generates grid data indicating a two-dimensional distribution of each of the trunks and the leaves of the rows of trees. The ECU **170** in this example determines a path for the tractor **100A** based on grid data as shown in FIG. **30**. For example, the ECU **170** sets a cell (3) that is located between two adjacent rows of trees as a travelable region. The ECU **180** controls the travel of the tractor **100A** so as to pass near the central portion of the travelable region.

(238) As described earlier, when generating an environment map through self-driving, the tractor

100A according to the present preferred embodiment collects local map data indicating a distribution of the trunks of rows of trees, while performing localization and steering control. At this time, based on the sensor data that is repeatedly output from the LiDAR sensor **110** while the tractor **100A** is moving, the ECU **160** may further detect the leaves of the rows of trees in the surrounding environment of the tractor **100A**, and generate data indicating a distribution of the detected leaves of the rows of trees. In that case, based on a distribution of each of the detected trunks and the detected leaves of the rows of trees, the ECU **180** can perform steering control of the tractor **100A**. The ECU **180** performs steering control of the tractor **100A** so that the tractor **100A** moves along a path which goes between two adjacent rows of trees among the detected rows of trees and which reduces contact with the leaves of the rows of trees, for example.

(239) In this operation, from the distribution of the trunks of the rows of trees as detected based on sensor data from the LiDAR sensor **110**, the ECU **170** determines a path which passes between the trunks of two adjacent rows of trees. For example, it determines a path which passes through the midpoint between the positions of the pair of trunks of two adjacent rows of trees. The ECU **180** performs steering control of the tractor **100A** so that the tractor **100A** moves along the determined path. As a result, self-driving can be performed while collecting local map data.

(240) In the environment in which the tractor **100A** travels, rows of trees may not necessarily be in a linear placement. Rows of trees may be in a curved placement. In such cases, when performing self-driving while collecting map data, a row of trees may be detected only on one of the right and left sides. In such cases, because right-and-left pairs of rows of trees cannot be identified, it is difficult to continue self-driving with the aforementioned method.

(241) Therefore, the following control may be performed in the case where rows of trees are in a curved placement. When rows of trees are in a curved placement, fewer trunks will be detected in one of any two adjacent rows of trees than in the other one of the two rows of trees. In such cases, based on a distribution of the detected trunks, the ECU **170** estimates a distribution of hidden trunks in the aforementioned one of the two rows of trees, and based on the estimated distribution, determines a path for the tractor **100A**. The ECU **180** performs steering control of the tractor **100A** so that the tractor **100A** travels along the determined path.

(242) FIG. **31A** is a diagram schematically showing an example environment in which rows of trees are in a curved placement. Columns in the figure represent trunks of trees. A thick broken line represents a schematic arrangement of a row of trees, and dotted lines schematically represent scans made by multiple layers of laser beams emitted from the LiDAR sensor **110**. FIG. **31B** is a diagram schematically a point cloud that may be observed through a single scan in the environment shown in FIG. **31A**. In this example, regarding the two adjacent rows of trees, three trunks **22R1**, **22R2** and **22R3** are detected in the right row of trees, whereas only one trunk **22L1** is detected in the left row of trees. In the left row of trees, the trunk **22L1** of a forward tree obstructs the laser beams so that the trunks of any rearward trees cannot be detected.

(243) In such a case, from the arrangement of the right row of trees, in which relatively many trees are detected, the ECU **170** estimates the positions of the trunks of the hidden left trees, and determines a travel path based on the estimated trunk positions. For example, the positions of the hidden trunks are estimated by applying to the two rearward adjacent trunks the relative position relationship of the two forward adjacent trunks **22R1** and **22L1**, whose positions are identified based on the scan data.

(244) FIG. **31C** is a diagram for describing an operation of determining a travel path by estimating the positions of hidden trunks. In this example, on the basis of trunk distribution data indicating a two-dimensional distribution of the trunks of rows of trees that is acquired based on scan data, positions of hidden trunks are estimated. In FIG. **31C**, positions of the trunks of trees which have been detected based on scan data are indicated with solid-line circles, whereas positions of the trunks of trees which are yet to be detected are indicated with broken-line circles. In this example, the ECU **170** first determines a vector **V1** from the position of the trunk **22R1** to the position of the

trunk **22L1**. Next, beginning from the position of the rearward trunk **22R2**, a position that is displaced by the vector **V1** from that position is determined as the estimated position of a hidden trunk. In FIG. **31C**, the estimated position of the hidden trunk is indicated by an x symbol. The ECU **170** determines a path which passes through a midpoint **C1** between that position and the position of the trunk **22R2** as a travel path for the tractor **100A**, for example.

(245) When the tractor **100A** travels along a path that is determined by such a method, a rearward trunk **22L2** that was hidden by the trunk **22L1** is eventually detected from scan data. It is further assumed that a rearward trunk **22L3** is yet to be detected. If the position of the trunk **22L2** is different from the estimated position (x symbol), the ECU **170** identifies a right trunk (i.e., the trunk **22R3** in the example of FIG. **31C**) that forms a pair with the trunk **22L2**, and determines a vector **V2** from that trunk **22R3** to the trunk **22L2**. Next, further beginning from the position of the rearward trunk **22R4**, a position that is displaced by the vector **V2** from that position is determined as the estimated position of the hidden trunk **22L3**. By repeating the above operation, positions of hidden trunks can be estimated to set an appropriate path, and automatic steering can be performed along that path.

(246) FIG. **31D** is a diagram showing an example of grid data indicating a two-dimensional distribution of trunks in the case where rows of trees are in a curved placement. The ECU **160** may generate grid data as shown in FIG. **31D** in the course of collecting data for establishing an environment map. In that case, the ECU **170** may perform path setting based on such grid data. In this example, positions of hidden trunks may be estimated in units equal to the cell size.

(247) Through the above operation, relatively smooth steering control can be performed while generating an environment map.

(248) Next, an obstacle avoidance operation according to the present preferred embodiment will be described.

(249) FIG. **32** is a diagram schematically showing an example situation where an obstacle **40** and a person **50** exist in the traveling environment of the tractor **100A**. The tractor **100A** includes one or more obstacle sensors **130**. During travel, if the obstacle sensor(s) **130** detects the obstacle **40** (e.g., a basket) or the person **50**, the ECU **180** performs steering control so as to avoid the obstacle **40** or the person **50**. In the case where collision with the obstacle **40**, the person **50**, branches, trunks **22**, etc., is inevitable even if avoiding attempts are made, the ECU **180** halts the tractor **100A**. Note that, when an obstacle is detected, the ECU **180** may halt the tractor **100A** anyway even if the obstacle is avoidable. Rather than using the obstacle sensor(s) **130**, the LiDAR sensor **110** may be used to perform similar detections.

(250) In the example shown in FIG. **32**, the person **50** exists near the trunk **22** of a tree. In the case where matching is performed in a manner that also encompasses the tree leaves, the person **50** or other obstacle may be hidden in the leaves, thus making it difficult to distinguish whether it is the leaves or any other object. In particular, distinction is difficult if the person **50** stands in the leaves near the trunk **22**. In the present preferred embodiment, because matching is performed by using an environment map indicating a distribution of trunks **22**, distinction between the trunks and other objects can be facilitated. An object other than a trunk that has been detected through matching between the scan data and the environment map can be easily detected to be a person or an obstacle. Thus, the LiDAR sensor **110** can be utilized as an obstacle sensor(s). In the case where the tractor **100A** includes a plurality of LiDAR sensors **110** at different positions of the tractor **100A**, the obstacle sensor(s) **130** may be omitted.

(251) Although the above preferred embodiment assumes that the interval between the trunks of trees is roughly constant, in practice, the trunks of trees may have different intervals in different places. Therefore, when performing localization, the ECU **160** may detect the trunks of multiple trees having a specific combination of trunk intervals from the sensor data, and estimate the position of the mobile machine through matching between the trunks of multiple trees having the specific combination of trunk intervals and the trunks of multiple trees having the specific

combination of trunk intervals as extracted from the environment map data. Hereinafter, this operation will be described with reference to FIGS. **33A** and **33B**.

(252) FIG. **33A** is a diagram showing an example placement of the trunks **22** of rows of trees detected by the tractor **100A**. FIG. **33B** is a diagram showing an example trunk distribution indicated by the environment map data. In this example, three trunks within a dotted-line box have a specific combination of trunk intervals of about 1.2 m and about 1.5 m. In such a case, when performing localization, the ECU **160** calculates respective intervals of the trunks **22** detected from the acquired scan data, and determines a portion having the specific combination of trunk intervals. Then, from the environment map shown in FIG. **33B**, it searches for and identifies a portion having the determined combination of trunk intervals. Through matching between the portions having the specific trunk intervals against each other, the ECU **160** is able to estimate the current position and orientation of the tractor **100A**. Through this operation, the matching can be made rapid and the localization accuracy can be improved in the case where the rows of trees include a portion having a specific combination of trunk intervals. Information indicating combinations of trunk intervals in the rows of trees may be previously recorded to a storage medium. Such information may be automatically recorded by the ECU **160** based on the environment map data, or recorded by the user manipulating the operation terminal **200**, for example.

(253) In the present preferred embodiment, the environment map data generated by the tractor **100A** can be supplied to other tractors, or mobile machines other than tractors, that are traveling in the same environment. For example, in the case where tasks are to be performed by a plurality of mobile machines in a broad orchard, it is sufficient if the environment map is generated by a single mobile machine or computer, and it will be efficient if the environment map is shared among the plurality of mobile machines. Local maps for establishing an environment map in a single environment may be generated by a plurality of tractors or other mobile machines in a shared fashion. In that case, a computer that combines the local maps generated by the plurality of mobile machines to generate a single environment map is to be provided within the system. That computer may deliver the generated environment map to the plurality of mobile machines through a wired or wireless network or a storage medium.

(254) In a system where a plurality of mobile machines work while moving in the same environment, the position relationship between the sensor coordinate system and the mobile machine coordinate system may differ from mobile machine to mobile machine. Even among the same models, errors in the attached positions of sensors may induce errors in the conversion from the sensor coordinate system to the mobile machine coordinate system. In order to reduce the influences of such errors, before beginning usual operations, each mobile machine makes a trial run to perform a calibration of determining parameters for the coordinate transformation from the sensor coordinate system to the mobile machine coordinate system.

(255) The techniques according to the present preferred embodiment and modifications and combinations thereof are applicable not only to tractors traveling in an orchard such as a vineyard, but also to any arbitrary mobile machine (e.g., a mobile robot or a drone) to be used in an environment where multiple rows of trees exist, e.g., a forest. The same is true of the following preferred embodiments.

Preferred Embodiment 2

(256) Next, a mobile machine according to a second preferred embodiment of the present disclosure will be described.

(257) The mobile machine according to the present preferred embodiment includes at least two LiDAR sensors, and performs localization and map data generation based on sensor data that is output from these LiDAR sensors. Each of the at least two LiDAR sensors outputs two-dimensional or three-dimensional point cloud data, or distance distribution data, indicating a distribution of objects in the surrounding environment of the mobile machine. As in Preferred Embodiment 1, the mobile machine includes a storage to store environment map data indicating a

distribution of trunks of the multiple rows of trees, a localization processor, and a controller to control movement of the mobile machine in accordance with the position of the mobile machine estimated by the localization processor. Based on the point cloud data that is repeatedly output from the at least two sensors while the mobile machine is moving, the localization processor is configured or programmed to detect trunks of the rows of trees in the surrounding environment of the mobile machine, and estimate the position of the mobile machine through matching between the detected trunks of the rows of trees and the environment map data. The mobile machine further includes a data generator to generate environment map data, or local map data from which to establish environment map data. While performing localization based on the point cloud data that is repeatedly output from the at least two sensors while the mobile machine is moving, the data generator detects trunks of the rows of trees in the surrounding environment of the mobile machine, and generates local map data, from which to generate environment map data indicating a distribution of the detected trunks of the rows of trees, and stores the local map data to the storage. Also, by integrating the local map data that is repeatedly generated while the mobile machine is moving, the data generator may generate an environment map data in a world coordinate system, and record it to the storage.

(258) Hereinafter, with respect to an example where the mobile machine is a tractor that travels in an orchard such as a vineyard, the configuration and operation according to the present preferred embodiment will be described. In the following description, differences from Preferred Embodiment 1 will mainly be described, while description of any overlapping aspects will be omitted.

(259) FIG. 34 is a schematic diagram of a tractor 100B according to the present preferred embodiment as viewed in a lateral direction. The tractor 100B includes a first LiDAR sensor 110A at the body lower front, and a second LiDAR sensor 110B at the upper front of the cabin. Other aspects are similar to those of the configuration shown in FIG. 7. Although FIG. 34 does not show the linkage device 108 and the implement 300 illustrated in FIG. 7, these may also be used in the present preferred embodiment.

(260) Each of the first LiDAR sensor 110A and the second LiDAR sensor 110B in the present preferred embodiment is a two-dimensional LiDAR sensor. In other words, in the example shown in FIG. 9A, each of the LiDAR sensors 110A and 110B is equivalent to a LiDAR sensor 110 of the case where $N=1$, i.e., emits a laser beam in only the direction of one angle of elevation. Each of the LiDAR sensors 110A and 110B outputs data indicating the distance and direction toward each reflection point or two-dimensional point cloud data. The two-dimensional point cloud data includes information of two-dimensional coordinates of multiple reflection points expressed by the respective sensor coordinate systems of the LiDAR sensors 110A and 110B. By using 2 two-dimensional LiDARs, costs can be reduced as compared to the case of using a three-dimensional LiDAR.

(261) The first LiDAR sensor 110A is mounted at a position that is lower than an average height of the trunks of the rows of trees in the environment in which the tractor 100B travels. The second LiDAR sensor 110B is mounted at a position that is higher than the average height of the trunks of the rows of trees. The first LiDAR sensor 110A may be placed at a height of e.g., not less than about 10 cm and not more than about 150 cm, and in one example not less than about 15 cm and not more than about 100 cm, from the ground surface. The second LiDAR sensor 110B may be placed at a position (e.g., about 2 m from the ground surface) that is higher than about 150 cm from the ground surface, for example.

(262) Similarly to the LiDAR sensor 110 in Preferred Embodiment 1, the first LiDAR sensor 110A is placed so as to emit laser pulses frontward from the tractor 100A. The first LiDAR sensor 110A in the example of FIG. 34 emits laser pulse to be essentially parallel to the ground surface. When the tractor 100B is located on a flat ground surface, the angle between the outgoing direction of a laser pulse emitted from the first LiDAR sensor 110A and the ground surface may be set within a

range of e.g., ± 20 degrees, and in one example within a range of about ± 10 degrees.

(263) On the other hand, the second LiDAR sensor **110B** is placed so as to emit laser pulses frontward and obliquely downward from the tractor **100B**. The second LiDAR sensor **110B** in the example of FIG. **34** emits laser pulse in a direction that is inclined by about 25 degrees from the ground surface. When the tractor **100B** is located on a flat ground surface, the angle between the outgoing direction of a laser pulse emitted from the second LiDAR sensor **110B** and the ground surface may be set within a range of e.g., not less than about 10 degrees and not more than about 45 degrees.

(264) FIG. **35** is a block diagram showing an exemplary schematic configuration of the tractor **100B** according to the present preferred embodiment. The exemplary configuration shown in FIG. **35** differs from the configuration shown in FIG. **8** in that the tractor **100B** includes the two LiDAR sensors **110A** and **110B**.

(265) The ECU **160** functions as the aforementioned localization processor and data generator. From the point cloud data that is repeatedly output from the two LiDAR sensors **110A** and **110B**, the ECU **160** acquires input point clouds from multiple scans including the latest scan, and performs matching between the input point clouds from multiple scans and the environment map data. As a result, the position and orientation of the tractor **100B** during travel can be estimated. Note that the ECU **160** may only estimate the position of the tractor **100B** through matching, and determine the orientation based on the signals output from the IMU **125**.

(266) In the course of establishing the environment map, based on the scan data that is repeatedly output from the two LiDAR sensors **110A** and **110B** while performing localization based on tractor **100B**, the ECU **160** detects trunks of the rows of trees in the surrounding environment of the mobile machine, and generates local map data, from which to generate environment map data indicating a distribution of the detected trunks of the rows of trees, and stores the local map data to the storage device **150**. While this operation is being performed, the ECU **170** sets a target path which passes between the trunks of the rows of trees as detected based on the scan data. By controlling the drive device **140** in accordance with the target path having been set, the ECU **180** causes the tractor **100B** to travel along the target path. Based on signals output from the obstacle sensor(s) **130** or scan data that is output from one or both of the LiDAR sensors **110A** and **110B**, the ECU **170** may detect obstacles, and set a target path so as to avoid contact with the obstacles. By joining together repeatedly generated local map data, the ECU **160** generates environment map data and records it to the storage device **150**. Detection of trunks of the rows of trees may be performed not at the stage of generating local map data, but at the stage of generating environment map data. The process of generating environment map data based on local map data may be performed by an external computer.

(267) During the operation of self-driving or automatic steering after the environment map has been established, the ECU **160** detects trunks of the rows of trees based on scan data that is repeatedly output from the LiDAR sensors **110A** and **110B**, and performs matching between the detected trunks and the trunks indicated by the environment map data, thus performing localization. The ECU **180** causes the tractor **100B** to travel along an intended travel path that is previously determined by the ECU **170**. In this case, too, the ECU **170** may detect obstacles based on signals that are output from the obstacle sensor(s) **130** during travel of the tractor **100B**, or scan data that is output from one or both of the LiDAR sensors **110A** and **110B**, and change the intended travel path so as to avoid contact with the obstacles.

(268) FIGS. **36A** to **36C** are diagrams schematically showing the tractor **100B** traveling near a tree **21**. The tractor **100B** gets closer to the tree **21** in the order of FIGS. **36A** to **36C**. In these figures, broken lines represent laser beams that are emitted from the LiDAR sensors **110A** and **110B**. As shown in the figures, in accordance with the distance between the tractor **100B** and the tree **21**, the position of the irradiated point on the trunk **22** of the tree **21** that is irradiated with a laser beam emitted from the second LiDAR sensor **110B** changes in the height direction. On the other hand,

the change in the position of the irradiated point on the trunk **22** of the tree **21** that is irradiated with a laser beam emitted from the first LiDAR sensor **110A** is relatively small. By combining the point cloud data that is acquired through multiple scans of these laser beams, as in the case of using a three-dimensional LiDAR, point cloud data indicating a three-dimensional distribution of reflection points on the surface of the trunk **22** can be acquired.

(269) The ECU **160** in the present preferred embodiment may accumulate scan data from a predetermined number (e.g., 3 or 5) of scans output from each of the LiDAR sensors **110A** and **110B**, and perform matching between the accumulated scan data and the environment map data, thus performing localization. By doing so, a highly accurate localization substantially similar to what is attained by using a three-dimensional LiDAR becomes possible.

(270) While the tractor **100B** is traveling in order to acquire map data, the ECU **170** determines a local target path for the tractor **100B** based on scan data from several scans that was output from one or both of the LiDAR sensors **110A** and **110B**. For example, from the placement of a point cloud indicated by scan data from several scans that was output from one or both of the LiDAR sensors **110A** and **110B**, a center position between two adjacent trees can be determined, and a local target path for the tractor **100B** can be determined so as to pass through this center position. With such path setting, the tractor **100B** is able to autonomously travel while maintaining equal or substantially equal distances to the right and left rows of trees.

(271) The trunk detection may be performed based only on the scan data that is output from the first LiDAR sensor **110A**. The scan data that is output from the first LiDAR sensor **110A**, which is at a relatively low position, is likely to include a large amount of information of a point cloud indicating the surface of a trunk. Therefore, distribution information of the trunks can still be acquired by using only the scan data from the first LiDAR sensor **110A**.

(272) Because the second LiDAR sensor **110B** in the present preferred embodiment is placed at a relatively high position, it is suitable for detection of leaves of the rows of trees. Therefore, the ECU **160** may generate data indicating a distribution of leaves of the rows of trees based on the point cloud data that is acquired with the second LiDAR sensor **110B**, and record it to the storage device **150** or another storage medium. Such data may be used for growth status management of the trees (e.g., canopy management).

(273) During travel of the tractor **100B**, obstacles existing at relatively high positions (e.g., leaves of the rows of trees) may be detected based on point cloud data that is acquired with the second LiDAR sensor **110B**, and an avoiding or halt operation may be performed. In the case where no obstacle sensor **130** is provided at the upper front of the tractor **100B**, if it were not for the second LiDAR sensor **110B** it would be difficult to detect and avoid leaves or vines of trees protruding above the path between rows of trees. Providing the second LiDAR sensor **110B** makes detection of such leaves or vines easier.

(274) According to the present preferred embodiment, by using 2 two-dimensional LiDARs, a similar operation to that of Preferred Embodiment 1 can be realized at low cost. Without being limited to 2, it is possible to provide 3 or more two-dimensional LiDARs at different positions. Moreover, one or more two-dimensional LiDARs and one or more three-dimensional LiDARs may be used in combination. Thus, the configuration according to the present preferred embodiment admits of various modifications. Note that the various techniques described with reference to Preferred Embodiment 1 are readily applicable to the present preferred embodiment.

Preferred Embodiment 3

(275) Next, a third preferred embodiment of the present disclosure will be described.

(276) The present preferred embodiment relates to a system that uses one or more mobile machines (e.g., a drone(s)) to collect local map data from which to establish environment map data, and generates environment map data based on the collected local map data.

(277) FIG. **37** is a diagram schematically showing an exemplary configuration of the system according to the present preferred embodiment. This system includes one or more drones **400**, one

or more tractors **100C**, and a server computer **500** (hereinafter referred to as the “server **500**”). These component elements are connected so as to be capable of communicating with each other via a wired or wireless network **60**. Although FIG. **37** illustrates two drones **400** and three tractors **100C**, the number of drones **400** and the number of tractors **100C** may be arbitrary. Each drone **400** generates local map data from which to establish environment map data while flying in an environment to be traveled by the tractors **100C**, and transmits it to the server **500**. Based on the local map data received from the one or more drones **400**, the server **500** generates environment map data that is expressed by a unified world coordinate system, and delivers it to each tractor **100C**. Based on the delivered environment map data, each tractor **100C** autonomously travels in the orchard.

(278) FIG. **38** is a block diagram showing an exemplary configuration of this system. The tractor **100C** in this example includes similar component elements to those of the tractor **100A** according to Preferred Embodiment 1 shown in FIG. **8**. However, the ECU **160** in the present preferred embodiment does not include a map data generating module **164**. The communication I/F **190** communicates with the server **500** via the network **60**. The storage device **150** stores the environment map data that is delivered from the server **500**. Note that an implement to be linked to the tractor **100C** in use is omitted from illustration in FIG. **38**. Similarly to Preferred Embodiment 1, the communication I/F **190** may perform communication with an implement that is connected to the tractor **100C**.

(279) Each drone **400** includes a data generation unit **450**, a drive device **440**, a controller **480**, and a communication I/F **490**. The drive device **440** includes various devices, such as an electric motor for driving purposes, multiple propellers, etc., which are necessary for the flight of the drone **400**. The controller **480** controls the operation of the data generation unit **450** and the drive device **440**. The communication I/F **490** is a circuit to communicate with the server **500** via the network **60**. The data generation unit **450** includes a LiDAR sensor **410** and a data generator **460**. The LiDAR sensor **410** has similar functionality to that of the LiDAR sensor **110** in Preferred Embodiment 1. The data generator **460** has similar functionality to that of the ECU **160** in Preferred Embodiment 1. In other words, the data generator **460** has the function of simultaneously performing localization and map generation. However, the data generator **460** does not generate the final environment map data. The data generator **460** includes a processor and a storage medium, e.g., a memory. The data generator **460** repeatedly generate local map data indicating a distribution of trunks of the multiple rows of trees based on scan data that is repeatedly output from the LiDAR sensor **410** during flight of the drone **400**, and accumulates it to the storage medium. The accumulated local map data is transmitted from the communication I/F **490** to the server **500** via manipulations by the user, for example.

(280) The server **500** may be a computer, e.g., a cloud server or an edge server, that is installed at a remote place from the tractors **100C** and the drones **400**, for example. The server **500** includes a storage device **550**, a processing device **560**, and a communication I/F **590**. The communication I/F **590** is a circuit for communicating with the tractors **100C** and the drones **400** via the network **60**. By unifying and integrating the coordinate systems of the local map data acquired from the multiple drones **400**, the processing device **560** generates environment map data, and records it to the storage device **550**. The processing device **560** delivers the generated environment map data from the communication I/F **590** to the multiple tractors **100C**. The processing device **560** may deliver the environment map data to mobile machines other than the tractors **100C**. For example, in the case where the drones **400** are to perform not only map data collection but also tasks such as seeding, manure spreading, or preventive pest control, the environment map data may be delivered to the drones **400**. In that case, the drones **400** can autonomously fly while performing localization through matching between the scan data that is output from the LiDAR sensor **410** and the environment map data, and perform predetermined tasks.

(281) FIG. **39** is a perspective view showing an example appearance of a drone **400**. This drone

400 includes a data generation unit **450** at the front surface of its body. The data generation unit **450** includes the LiDAR sensor **410**, and is able to scan the surrounding environment with laser beams while flying.

(282) FIG. **40** is a perspective view schematically an example manner of data collecting work by multiple drones **400**. FIG. **40** illustrates two drones **400** flying between multiple rows of trees **20**. While autonomously moving between the rows of trees **20** through low-altitude flight, the drones **400** repeat the operation of scanning the trunks of the rows of trees **20** with laser beams to generate local map data indicating a distribution of the trunks. The method of generating the local map data is similar to the method in Preferred Embodiment 1.

(283) FIG. **41** is a diagram schematically showing a tree **21** being irradiated with laser beams emitted from the LiDAR sensor **410** mounted on a drone **400**. Each drone **400** in the present preferred embodiment is programmed to fly at an altitude such that the LiDAR sensor **410** keeps a position that is lower than an average height of the trunks of the rows of trees. The altitude of the drone **400** is controlled so that the LiDAR sensor **410** is located at a height of not less than about 15 cm and not more than about 100 cm from the ground surface, for example. While performing localization based on sensor data that is repeatedly output from the LiDAR sensor **410** located at such a low altitude, the data generator **460** detects the trunks of the rows of trees in the surrounding environment of the drone **400**, generates local map data indicating a distribution of the detected trunks of the rows of trees, and records it to the storage medium. From the sensor data that is repeatedly output from the LiDAR sensor **410**, the data generator **460** acquires input point clouds from multiple scans performs matching between the input point cloud from at least one scan including the latest scan and the local map data generated in the previous run or earlier, thus performing localization.

(284) FIG. **42** is a diagram showing an example format of local map data to be output from each drone **400**. In this example, the local map data includes a mobile machine ID as an identification number of the drone **400**, the position (x,y,z) and attitude ($\theta_{\text{sub.R}}$, $\theta_{\text{sub.P}}$, $\theta_{\text{sub.Y}}$) of the drone **400** in a world coordinate system, and information that is recorded for each reflection point. The information to be recorded for each reflection point is similar to the information shown in FIG. **23**. However, the coordinates of each reflection point are coordinates (u,v,w) in the sensor coordinate system that is fixed to the LiDAR sensor **410** of the drone **400**. Such local map data may be transmitted from each drone **400** to the server **500**. Note that information concerning whether the given point is a trunk or not, e.g., classification and trunk ID, does not need to be included in the local map data. After acquiring the local map data from the multiple drones **400**, the processing device **560** in the server **500** may detect point clouds corresponding to the trunks based on characteristic aspects of the distribution of point clouds.

(285) With the above configuration, even in a broad orchard, for example, data for establishing the environment map can be efficiently collected.

(286) In the present preferred embodiment, not only the drones **400**, but also other mobile machines such as the tractors **100C** may also perform the operation of acquiring local map data from which to establish the environment map data. Moreover, the mobile machine to generate the local map data does not need to be a mobile machine that is capable of moving autonomously. For example, the local map data may be generated as the user drives or operates a mobile machine, e.g., a tractor or a drone, that includes one or more LiDAR sensors mounted at a position lower than an average height of the trunks of rows of trees.

(287) Without providing the server **500**, a mobile machine such as a drone **400** or a tractor **100C** may generate the final environment map data, and supply the environment map data to the other mobile machines. In that case, the environment map data is directly exchanged through communication between mobile machines.

3-4. Other Preferred Embodiments

(288) In the above preferred embodiments, each tractor may be an unmanned tractor. In that case,

component elements which are needed only for human driving, e.g., the cabin, the driver's seat, the steering wheel, and the operation terminal, may not be provided in the tractor. The unmanned tractor may perform a similar operation to the operation in each of the above-described preferred embodiments through autonomous driving, or remote manipulation by the user of the tractor.

(289) In the above preferred embodiments, the one or more sensors provided in the mobile machine is a LiDAR sensor that performs laser beam scanning in order to output two-dimensional or three-dimensional point cloud data, or distance distribution data, as the sensor data. However, the sensors are not limited to such LiDAR sensors. For example, a flash-type LiDAR sensor or other types of sensors, e.g., image sensors, may be used. Such other types of sensors may be combined with a scan-type LiDAR sensor.

(290) A device that performs the processing needed for the localization and autonomous movement (or automatic steering) or map data generation according to the above preferred embodiments may be mounted to a mobile machine lacking such functionality as an add-on. For example, a control unit to control the operation of a mobile machine that moves between multiple rows of trees may be attached to the mobile machine in use. Such a control unit includes one or more sensors to output sensor data indicating a distribution of objects in a surrounding environment of the mobile machine, a storage to store environment map data indicating a distribution of trunks of the multiple rows of trees, a localization processor, and a controller to control movement of the mobile machine in accordance with the position of the mobile machine estimated by the localization processor. Based on sensor data that is repeatedly output from one or more sensors while the mobile machine is moving, the localization processor is configured or programmed to detect trunks of the rows of trees in the surrounding environment of the mobile machine, and estimate the position of the mobile machine through matching between the detected trunks of the rows of trees and the environment map data. Also, a data generator to generate map data may be attached to a mobile machine that moves between multiple rows of trees in use. Such a data generator includes one or more sensors to output sensor data indicating a distribution of objects in a surrounding environment of the mobile machine, and a data generator. While performing localization based on sensor data that is repeatedly output from one or more sensors while the mobile machine is moving, the data generator is configured or programmed to detect trunks of the rows of trees in the surrounding environment of the mobile machine, and generate local map data, from which to generate environment map data indicating a distribution of the detected trunks of the rows of trees, and stores the local map data to the storage.

(291) As described above, the present disclosure encompasses mobile machines, control units, methods, and computer programs as recited in the following Items.

(292) Item 1

(293) A mobile machine movable between multiple rows of trees includes one or more sensors to output sensor data indicating a distribution of objects in a surrounding environment of the mobile machine, a storage to store environment map data indicating a distribution of trunks of the multiple rows of trees, a localization processor to detect the trunks of the rows of trees in the surrounding environment of the mobile machine based on the sensor data that is repeatedly output from the one or more sensors while the mobile machine is moving, and to perform matching between the detected trunks of the rows of trees and the environment map data to provide an estimated position of the mobile machine, and a controller to control movement of the mobile machine in accordance with the estimated position of the mobile machine.

(294) Item 2

(295) In the mobile machine described above, the one or more sensors include at least one LiDAR sensor to output two-dimensional or three-dimensional point cloud data as the sensor data.

(296) Item 3

(297) In the mobile machine described above, the at least one LiDAR sensor is mounted at a position that is lower than an average height of the trunks of the rows of trees.

(298) Item 4

(299) In the mobile machine described above, the at least one LiDAR sensor is at a height that is not less than about 15 cm and not more than about 100 cm from a ground surface.

(300) Item 5

(301) In the mobile machine described above, the localization processor is configured or programmed to acquire input point clouds from multiple scans from sensor data that is repeatedly output from the at least one LiDAR sensor mounted at a position that is lower than an average height of the trunks of the rows of trees, and to perform matching between the input point cloud from at least one scan including a latest scan and the environment map data to estimate the position of the mobile machine.

(302) Item 6

(303) In the mobile machine described above, the at least one LiDAR sensor mounted at a position that is lower than an average height of the trunks of the rows of trees includes a plurality of laser light sources to emit a laser pulse at a different angle of elevation, and the localization processor is configured or programmed to detect the trunks based on reflection points of the laser pulses emitted from, among the plurality of laser light sources, the laser light sources with an angle of elevation included in a predetermined range.

(304) Item 7

(305) In the mobile machine described above, the localization processor is configured or programmed to detect the trunks based on changes in distance from the at least one LiDAR sensor to reflection points of the laser pulses during scans by the at least one LiDAR sensor.

(306) Item 8

(307) In the mobile machine described above, the one or more sensors include at least one LiDAR sensor to output two-dimensional or three-dimensional point cloud data as the sensor data, the at least one LiDAR sensor is operable to repeatedly output the point cloud data with a pre-designated cycle, and the localization processor is configured or programmed to extract points representing a surface of the trunks based on the position of each point in the point cloud data that is output during a period of one cycle or longer or on the distance or angle of each point from the mobile machine, and to perform matching between the extracted points and the environment map data to estimate the position of the mobile machine.

(308) Item 9

(309) The mobile machine described above also includes an inertial measurement unit, wherein the localization processor is configured or programmed to estimate a position of the mobile machine by utilizing a signal that is output from the inertial measurement unit.

(310) Item 10

(311) In the mobile machine described above, the localization processor is configured or programmed to detect from the sensor data the trunks of multiple trees with a specific combination of trunk intervals, and to perform matching between the trunks of the multiple trees with the specific combination of trunk intervals and the trunks of the multiple trees with the specific combination of trunk intervals as extracted from the environment map data to estimate the position of the mobile machine.

(312) Item 11

(313) In the mobile machine described above, the controller is configured or programmed to determine whether an obstacle exists in a path of the mobile machine based on the environment map data and the sensor data, and when determining that the obstacle exists, cause the mobile machine to perform an operation of avoiding collision with the obstacle.

(314) Item 12

(315) The mobile machine described above further includes an obstacle sensor to detect an obstacle in a path of the mobile machine, resulting in the controller causing the mobile machine to perform an operation of avoiding collision with the obstacle.

(316) Item 13

(317) The mobile machine described above includes a calculator to measure a thickness of each detected trunk of the rows of trees based on the sensor data, and record the thickness of each trunk to a storage.

(318) Item 14

(319) In the mobile machine described above, one or more posts are provided in an area where the multiple rows of trees are placed, the environment map data indicates a distribution of the post or posts in addition to the distribution of the trunks, and the localization processor is configured or programmed to further detect the post or posts in the surrounding environment of the mobile machine based on the sensor data, and perform matching between the detected trunks of the rows of trees and the detected post or posts and the environment map data to estimate the position of the mobile machine.

(320) Item 15

(321) In the mobile machine described above, the localization processor is configured or programmed to detect the trunks of the rows of trees in the surrounding environment of the mobile machine based on the sensor data and on information indicating a placement relationship of the multiple rows of trees that is recorded on a storage.

(322) Item 16

(323) In the mobile machine described above, information indicating the placement relationship of the multiple rows of trees is recorded to the storage by being input via a Human Machine Interface.

(324) Item 17

(325) In the mobile machine described above, the mobile machine is a work vehicle that is capable of autonomously moving.

(326) Item 18

(327) In the mobile machine described above, the localization processor is configured or programmed to generate data indicating a distribution of the trunks of the rows of trees as detected based on the sensor data, and update the environment map data by using the data indicating the distribution of the trunks, or transmit the data indicating the distribution of the trunks to an external device that updates the environment map data.

(328) Item 19

(329) A control unit to control operation of a mobile machine movable between multiple rows of trees includes one or more sensors attached to the mobile machine in use to output sensor data indicating a distribution of objects in a surrounding environment of the mobile machine, a storage to store environment map data indicating a distribution of trunks of the multiple rows of trees, a localization processor to detect the trunks of the rows of trees in the surrounding environment of the mobile machine based on the sensor data that is repeatedly output from the one or more sensors while the mobile machine is moving, and to perform matching between the detected trunks of the rows of trees and the environment map data to provide an estimated position of the mobile machine, and a controller to control movement of the mobile machine in accordance with the estimated position of the mobile machine.

(330) Item 20

(331) A method of controlling operation of a mobile machine movable between multiple rows of trees includes acquiring environment map data indicating a distribution of trunks of the multiple rows of trees, while the mobile machine is moving, acquiring sensor data indicating a distribution of objects in a surrounding environment of the mobile machine from one or more sensors mounted to the mobile machine, detecting the trunks of the rows of trees in the surrounding environment of the mobile machine based on the acquired sensor data, performing matching between the detected trunks of the rows of trees and the environment map data to provide an estimated position of the mobile machine, and controlling movement of the mobile machine in accordance with the estimated position of the mobile machine.

(332) Item 21

(333) A non-transitory computer-readable medium includes a program to be executed by a computer to control operation of a mobile machine movable between multiple rows of trees, the computer program causing the computer to perform acquiring environment map data indicating a distribution of trunks of the multiple rows of trees, while the mobile machine is moving, acquiring sensor data indicating a distribution of objects in a surrounding environment of the mobile machine from one or more sensors mounted to the mobile machine, detecting the trunks of the rows of trees in the surrounding environment of the mobile machine based on the acquired sensor data, performing matching between the detected trunks of the rows of trees and the environment map data to provide an estimated position of the mobile machine, and controlling movement of the mobile machine in accordance with the estimated position of the mobile machine.

(334) The techniques according to preferred embodiments of the present disclosure and modifications or combinations thereof are applicable to tractors, drones, walking robots or other mobile machines that move in an environment where multiple rows of trees exist, e.g., a vineyard or any other orchard.

(335) While preferred embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

Claims

1. A mobile machine movable between multiple rows of trees, the mobile machine comprising: one or more sensors configured to output sensor data indicating a distribution of objects in a surrounding environment of the mobile machine; a storage configured to store environment map data indicating a distribution of trunks of the multiple rows of trees; a localization processor configured to detect the trunks of the rows of trees in the surrounding environment of the mobile machine based on the sensor data that is repeatedly output from the one or more sensors while the mobile machine is moving, and to perform matching between the detected trunks of the rows of trees and the environment map data to provide an estimated position of the mobile machine; and a controller configured to control movement of the mobile machine in accordance with the estimated position of the mobile machine; wherein the controller is configured to, when fewer trunks are detected in one of two adjacent rows of trees among the multiple rows of trees than in another one of the two rows of trees, estimate a distribution of undetected trunks in the one of the two rows of trees based on the distribution of the detected trunks, and perform steering control of the mobile machine based on the estimated distribution.

2. The mobile machine of claim 1, wherein the one or more sensors include at least one LiDAR sensor to output two-dimensional or three-dimensional point cloud data as the sensor data.

3. The mobile machine of claim 2, wherein the at least one LiDAR sensor is mounted at a position that is lower than an average height of the trunks of the rows of trees.

4. The mobile machine of claim 3, wherein the at least one LiDAR sensor is at a height that is not less than 15 cm and not more than 100 cm from a ground surface.

5. The mobile machine of claim 3, wherein the localization processor is configured to acquire input point clouds from multiple scans from sensor data that is repeatedly output from the at least one LiDAR sensor mounted at a position that is lower than an average height of the trunks of the rows of trees, and perform matching between the input point cloud from at least one scan including a latest scan and the environment map data to estimate a position of the mobile machine.

6. The mobile machine of claim 3, wherein the at least one LiDAR sensor mounted at a position that is lower than an average height of the trunks of the rows of trees includes a plurality of laser light sources to emit a laser pulse at a different angle of elevation; and the localization processor is

configured to detect the trunks based on reflection points of the laser pulses emitted from, among the plurality of laser light sources, the laser light sources with an angle of elevation within a predetermined range.

7. The mobile machine of claim 6, wherein the localization processor is configured to detect the trunks based on changes in distance from the at least one LiDAR sensor to reflection points of the laser pulses during scans by the at least one LiDAR sensor.

8. The mobile machine of claim 1, wherein, the one or more sensors include at least one LiDAR sensor to output two-dimensional or three-dimensional point cloud data as the sensor data; the at least one LiDAR sensor is operable to repeatedly output the point cloud data with a pre-designated cycle; and the localization processor is configured to extract points representing a surface of the trunks based on a position of each point in the point cloud data that is output during a period of one cycle or longer or based on a distance or an angle of each point from the mobile machine, and to perform matching between the extracted points and the environment map data to estimate a position of the mobile machine.

9. The mobile machine of claim 1, further comprising an inertial measurement unit, wherein the localization processor is configured to estimate a position of the mobile machine by utilizing a signal that is output from the inertial measurement unit.

10. The mobile machine of claim 1, wherein the localization processor is configured to detect from the sensor data the trunks of multiple trees having a specific combination of trunk intervals, and to perform matching between the trunks of the multiple trees having the specific combination of trunk intervals and the trunks of the multiple trees having the specific combination of trunk intervals as extracted from the environment map data to estimate the position of the mobile machine.

11. The mobile machine of claim 1, wherein the controller is configured to determine whether an obstacle exists in a path of the mobile machine based on the environment map data and the sensor data, and when determining that the obstacle exists, to cause the mobile machine to perform an operation of avoiding a collision with the obstacle.

12. The mobile machine of claim 1, further comprising an obstacle sensor, wherein when the obstacle sensor detects an obstacle in a path of the mobile machine, the controller is configured to cause the mobile machine to perform an operation of avoiding collision with the obstacle.

13. The mobile machine of claim 1, further comprising a calculator configured to measure a thickness of each detected trunk of the rows of trees based on the sensor data, and record the thickness of each trunk to a storage medium.

14. The mobile machine of claim 1, wherein the localization processor is configured to detect the trunks of the rows of trees in the surrounding environment of the mobile machine based on the sensor data and based on information indicating a placement relationship of the multiple rows of trees that is recorded on a storage medium.

15. The mobile machine of claim 1, wherein the mobile machine is a work vehicle that is capable of autonomously moving.

16. The mobile machine of claim 1, wherein the localization processor is configured to: generate data indicating a distribution of the trunks of the rows of trees as detected based on the sensor data; and update the environment map data by using the data indicating the distribution of the trunks, or transmit the data indicating the distribution of the trunks to an external device that updates the environment map data.

17. A control unit to control operation of a mobile machine movable between multiple rows of trees, the control unit comprising: one or more sensors attached to the mobile machine and configured to output sensor data indicating a distribution of objects in a surrounding environment of the mobile machine; a storage configured to store environment map data indicating a distribution of trunks of the multiple rows of trees; a localization processor configured to detect the trunks of the rows of trees in the surrounding environment of the mobile machine based on the sensor data that is repeatedly output from the one or more sensors while the mobile machine is moving, and to

perform matching between the detected trunks of the rows of trees and the environment map data to provide an estimated position of the mobile machine; and a controller configured to control movement of the mobile machine in accordance with the estimated position of the mobile machine; wherein the controller is configured to, when fewer trunks are detected in one of two adjacent rows of trees among the multiple rows of trees than in another one of the two rows of trees, estimate a distribution of undetected trunks in the one of the two rows of trees based on the distribution of the detected trunks, and perform steering control of the mobile machine based on the estimated distribution.

18. A method of controlling operation of a mobile machine movable between multiple rows of trees, the method comprising: acquiring environment map data indicating a distribution of trunks of the multiple rows of trees; while the mobile machine is moving, acquiring sensor data indicating a distribution of objects in a surrounding environment of the mobile machine from one or more sensors mounted to the mobile machine; detecting the trunks of the rows of trees in the surrounding environment of the mobile machine based on the acquired sensor data; performing matching between the detected trunks of the rows of trees and the environment map data to provide an estimated position of the mobile machine; and controlling movement of the mobile machine in accordance with the estimated position of the mobile machine; and when fewer trunks are detected in one of two adjacent rows of trees among the multiple rows of trees than in another one of the two rows of trees, estimating a distribution of undetected trunks in the one of the two rows of trees based on the distribution of the detected trunks, and performing steering control of the mobile machine based on the estimated distribution.
